Home-usage-power-detector

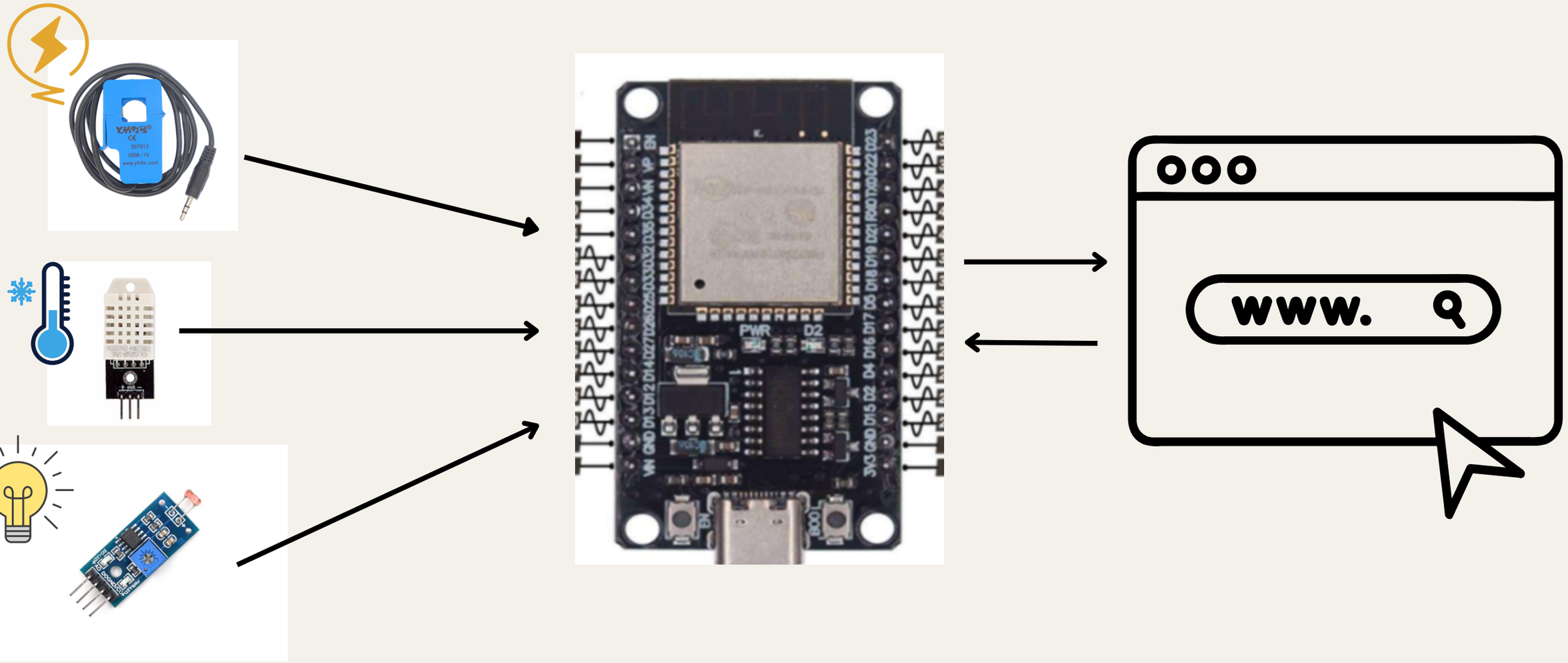
By G01 Siratee Tanjoy

REACH OUT

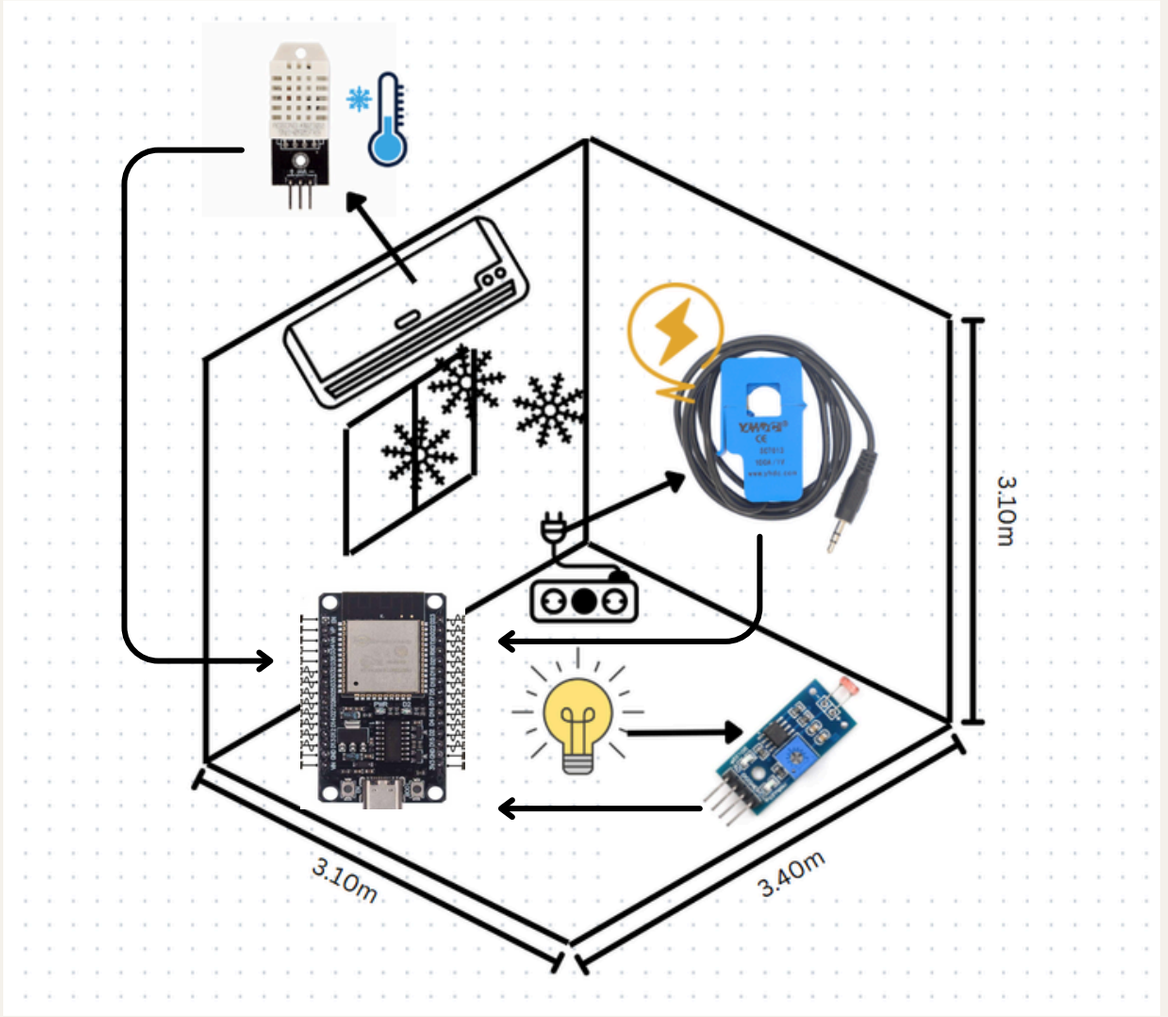
Project Requirement

- BCG or ESG ralate (Topic: Smart home usage power detector)
- new self-study (CT sensor, DHT22, LDR sensor, real time website)
- version control (GitHub)
- RTOS (Use Mutex to control task)

Overview



Overview Design



Detail/Requirement

Detail

- ใช้ LDR Photoresistor Sensor ตรวจจับความเข้มแสงเพื่อแสดงสถานะว่าภายในห้อง สว่าง หรือ มืด
- สามารถ เปิด-ปิด ไฟผ่านหน้าเว็บได้ (ver.จำลอง) และคำนวณค่าไฟตามช่วงที่เปิด
- ใช้ DHT22 ตรวจจับ อุณหภูมิ และ ควาามชื้อ เพื่อแสดงสถานะว่าภายในห้อง ร้อน หรือ เย็น
- สามารถ เปิด-ปิด เครื่องปรับอากาศผ่านหน้าเว็บได้ (ver.จำลอง) และคำนวณค่าไฟตามช่วงที่เปิด
- ใช้ CT Sensor ตรวจจับกระแสไฟไหลผ่านของสายไฟเพื่อแสดงคำนวณการใช้ไฟฟ้า และ ค่าไฟ

Requirement

- ห้องขนาดใดก็ได้ที่มี Internet เข้าถึง
- สายไฟที่อยู่ระหว่างแหล่งจ่ายไฟกับเครื่องใช้ไฟฟ้าต้องเป็นสายแบบ Single-Core ถ้าเป็นแบบ Multi-Core ต้องสามารถเข้าถึงสายต่างๆได้

Specification

Hardware

- ESP32 Dev Board CH340 x1
- Expansion Board ESP-32 WROOM 30pin GPIO TYPE-C USB x1
- LDR Photoresistor Sensor Module x1
- DHT22 AM2302 Module x1
- CT Sensor 100A SCT-013-000 x2
- 10uf Capacitor x2
- 10k Ohm Resistor x4
- 33 Ohm Resistor x2
- TRRS 3.5mm Jack Breakout x2 (Optional)
- Breadboard x1

Specification

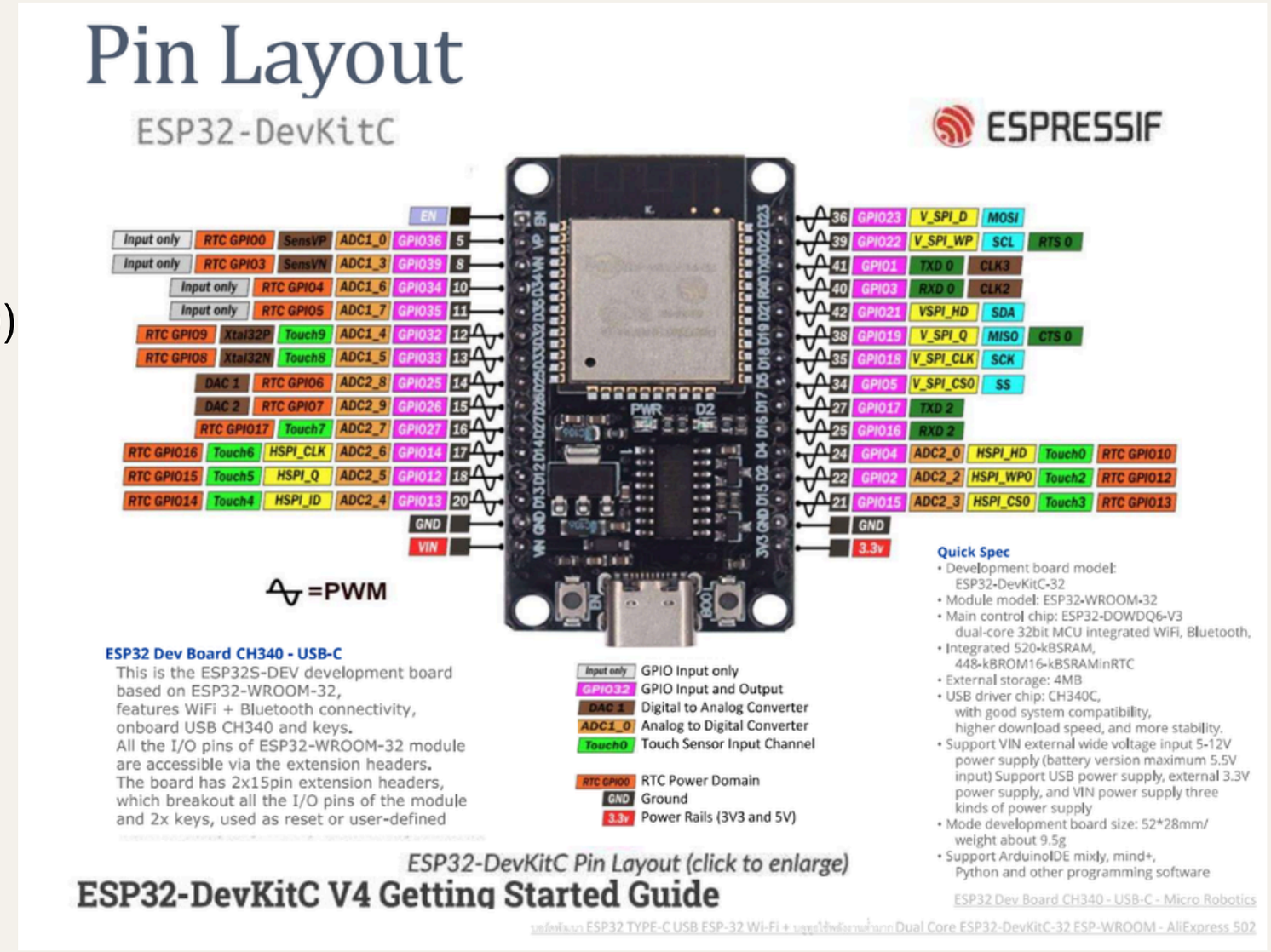
Software

- **Framework**: Arduino
- **Version control**: GitHub
- adafruit/DHT sensor library@^1.4.6
  #include <DHT.h>
  #include <Adafruit_Sensor.h>
- openenergymonitor/EmonLib@^1.1.0
  #include "EmonLib.h"
- esphome/ESPAsyncWebServer-esphome@^3.3.0
  #include <ESPAsyncWebServer.h>
- #include <Arduino.h>
- #include <WiFi.h>
- #include "index.h"
- task processing
- Mutex key

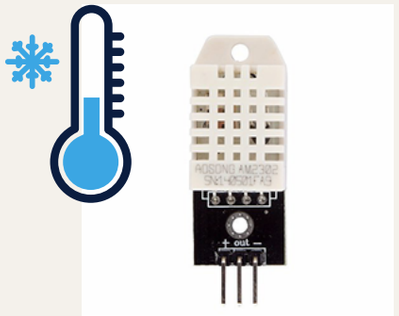
Architectural Design

- VN(GPIO39)
- GPIO35

GPIO34

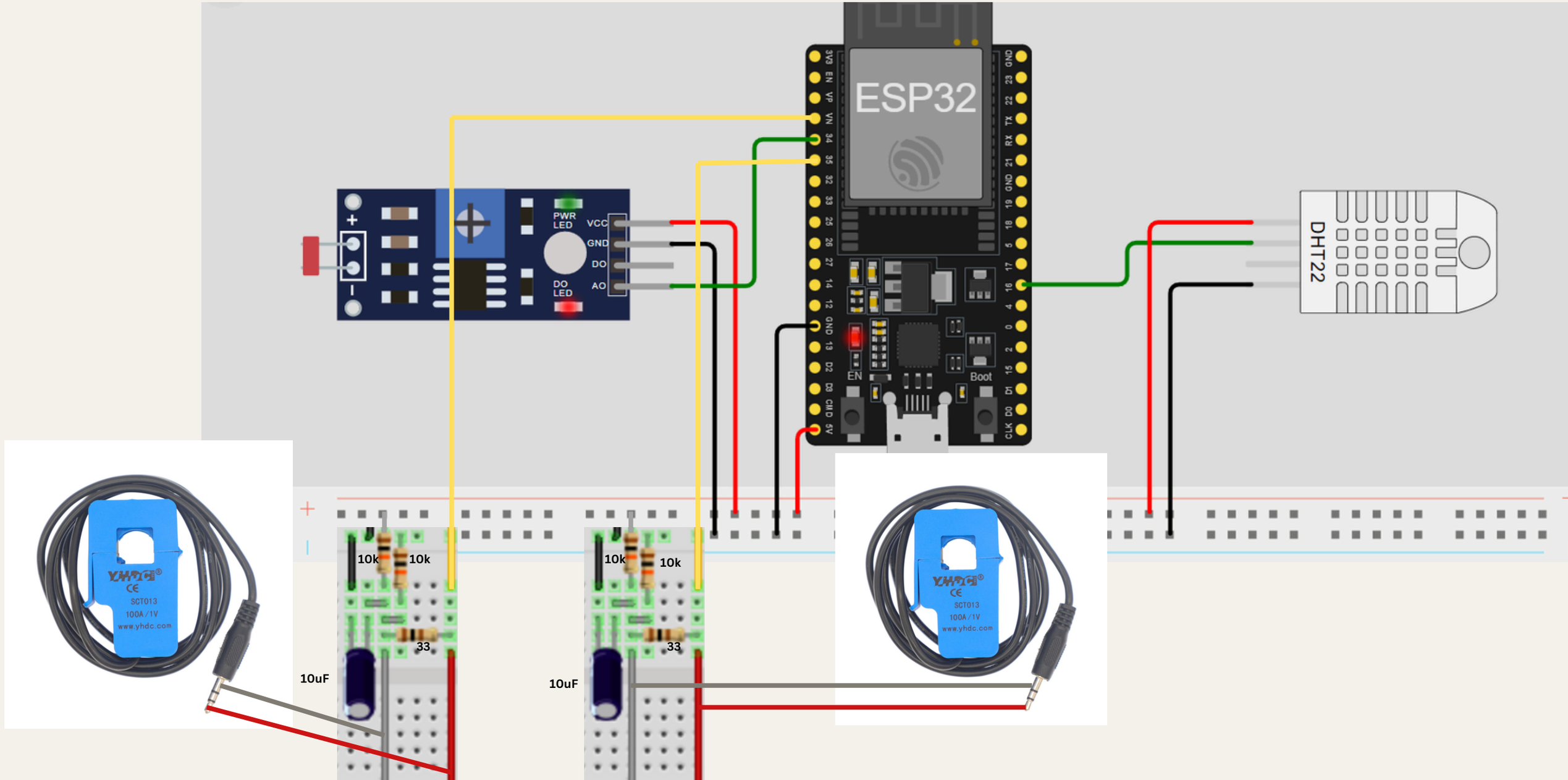


- VCC 3.3V or 5V
- GND

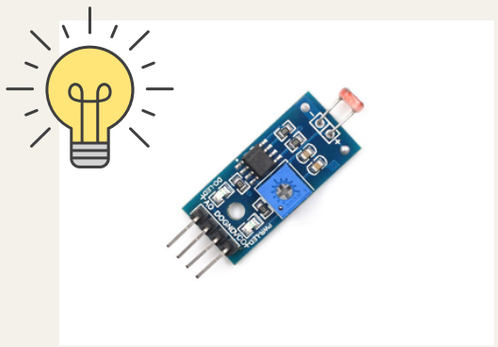


GPIO16

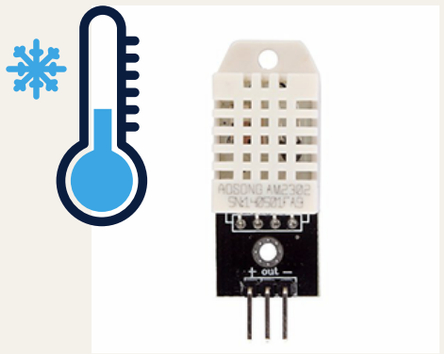
Architectural Design



Detailed Design



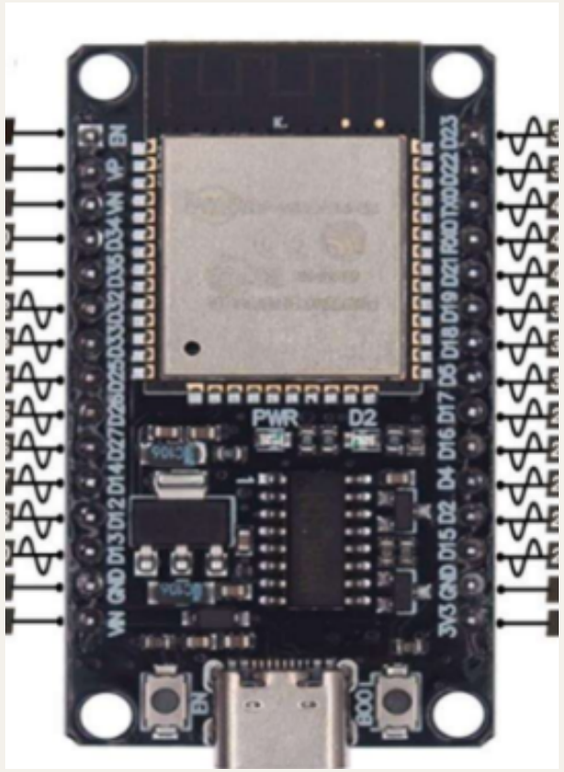
- get analog light value



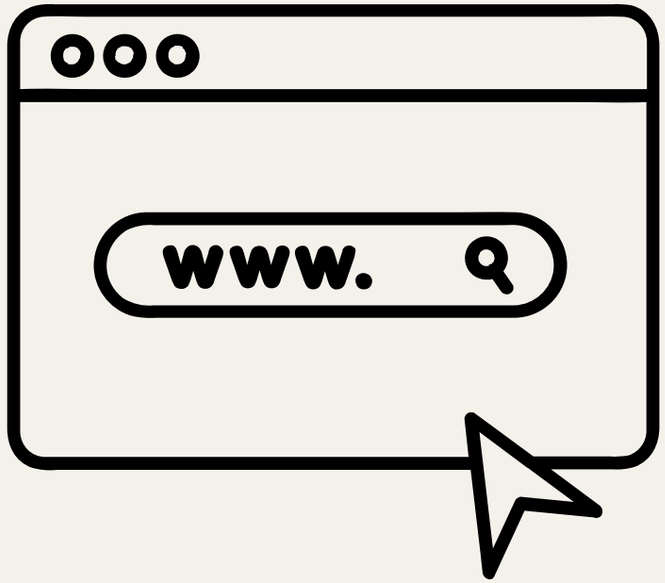
- get analog temperature value
- get analog humidlity value

- get mean of analog current

Detailed Design

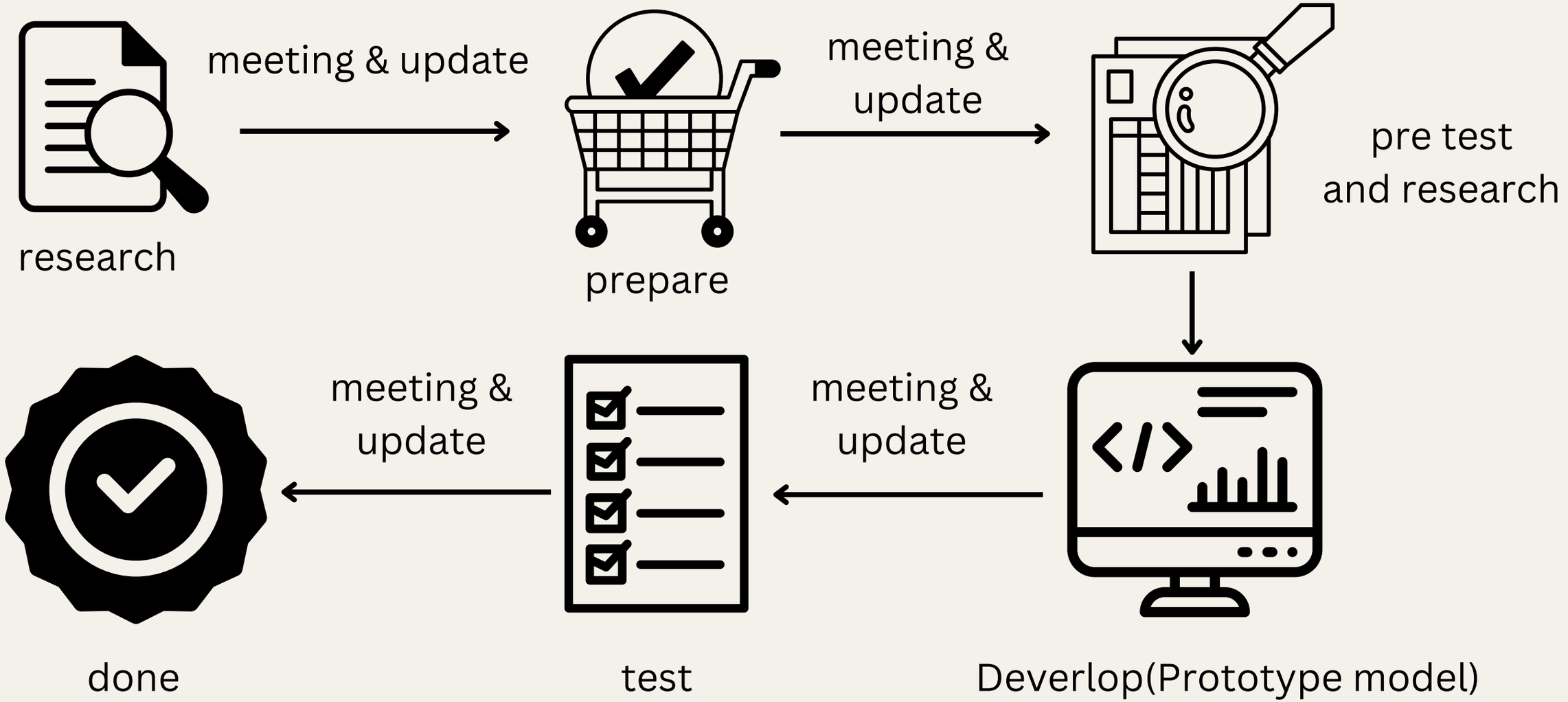


- get and calculate values from sensors
- send values to website
- host the website
- get control requests from website
- processes by tasks, each task has 22.5 secs delay

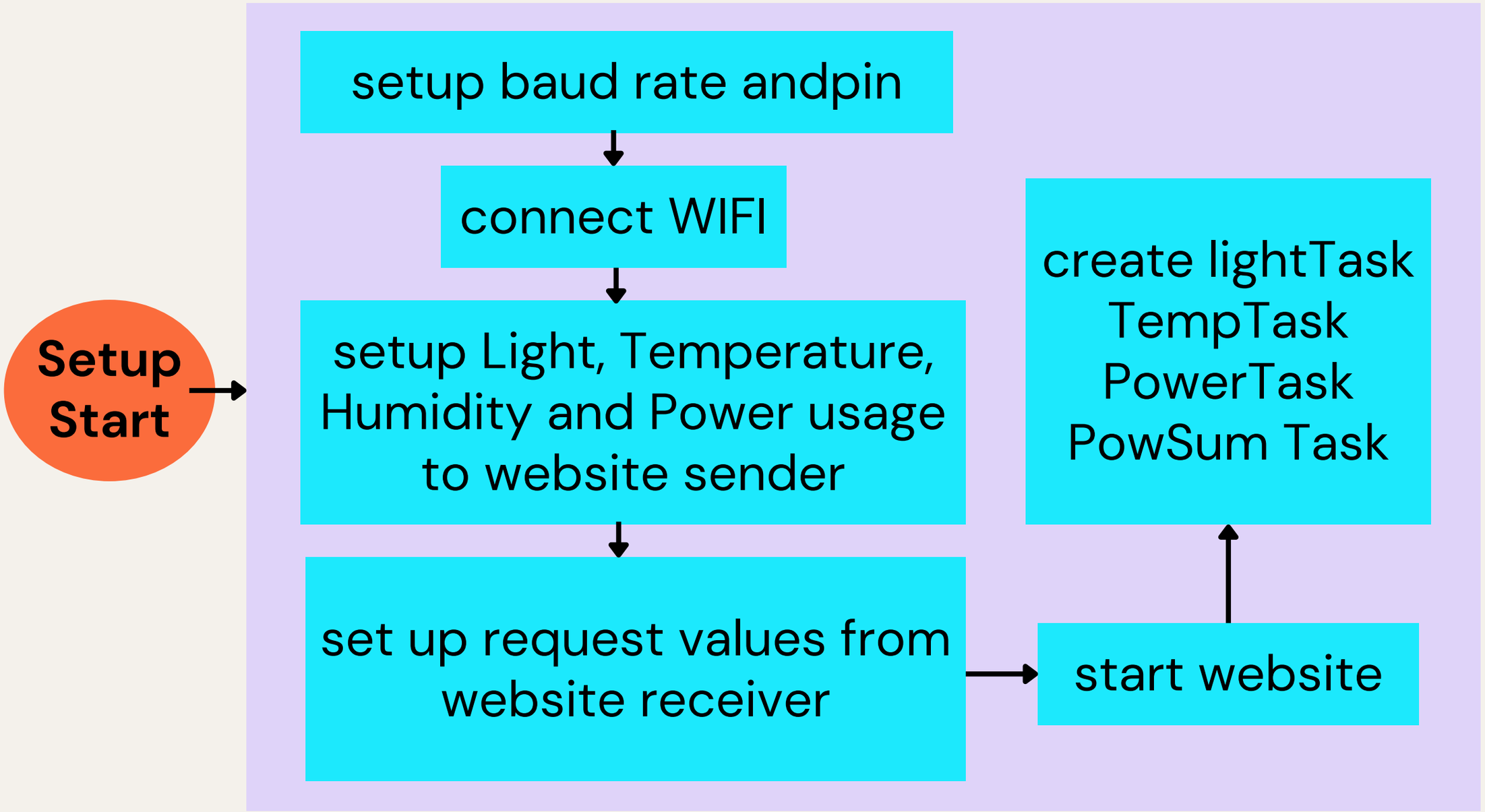


- get values from ESP32 and display
- sent control request
- refresh every 22.6 secs

Work Flow

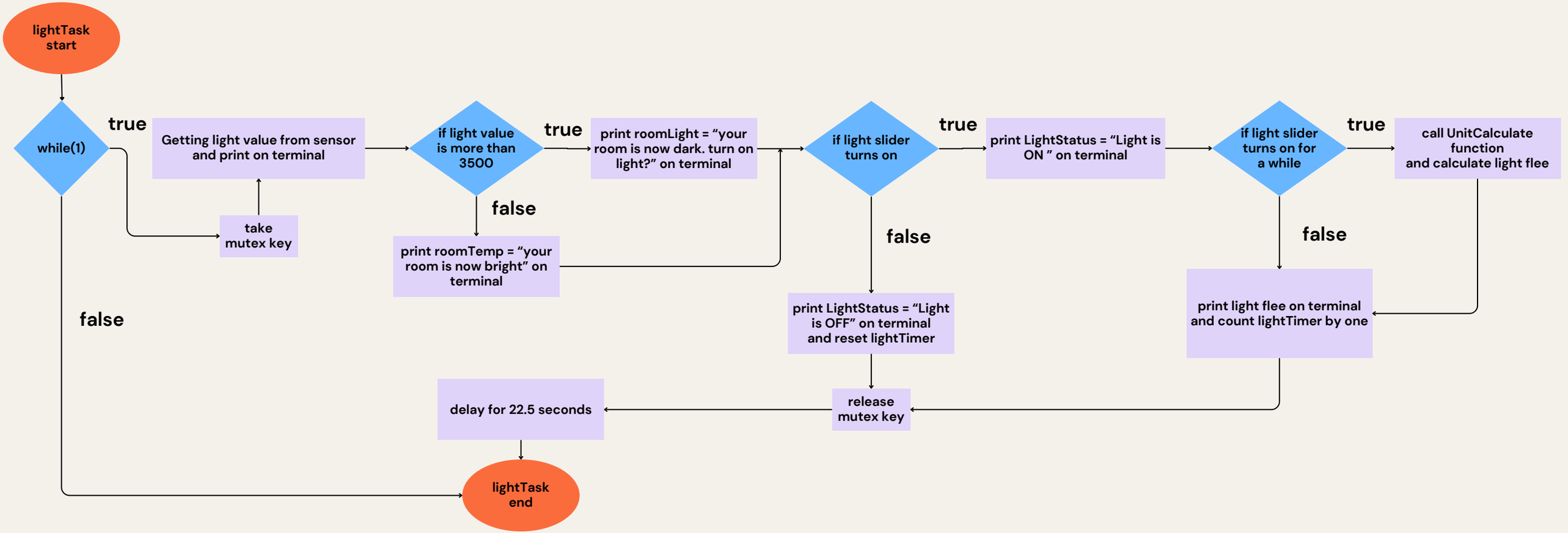

meeting & update
meeting & update
pre test and research
research
prepare
meeting & update
meeting & update
done
test
Deverlop(Prototype model)

Flow Chart


Setup Start
setup baud rate andpin
connect WIFI
setup Light, Temperature, Humidity and Power usage to website sender
set up request values from website receiver
start website
create lightTask
TempTask
PowerTask
PowSum Task

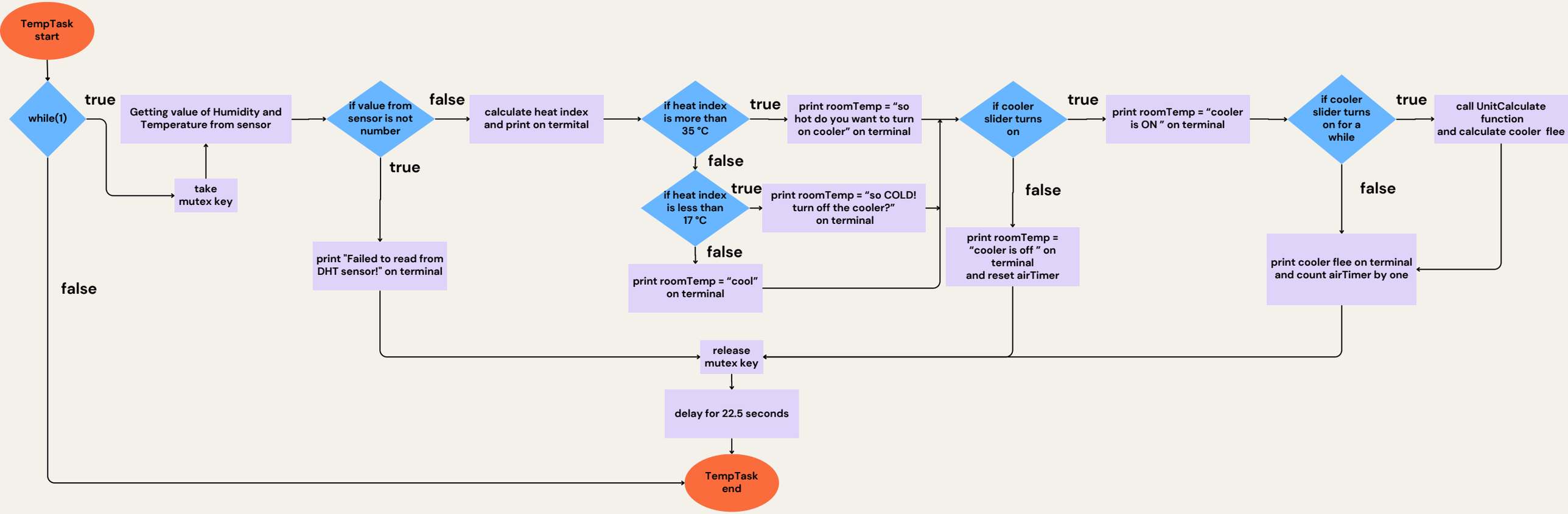
Flow Chart

lightTask



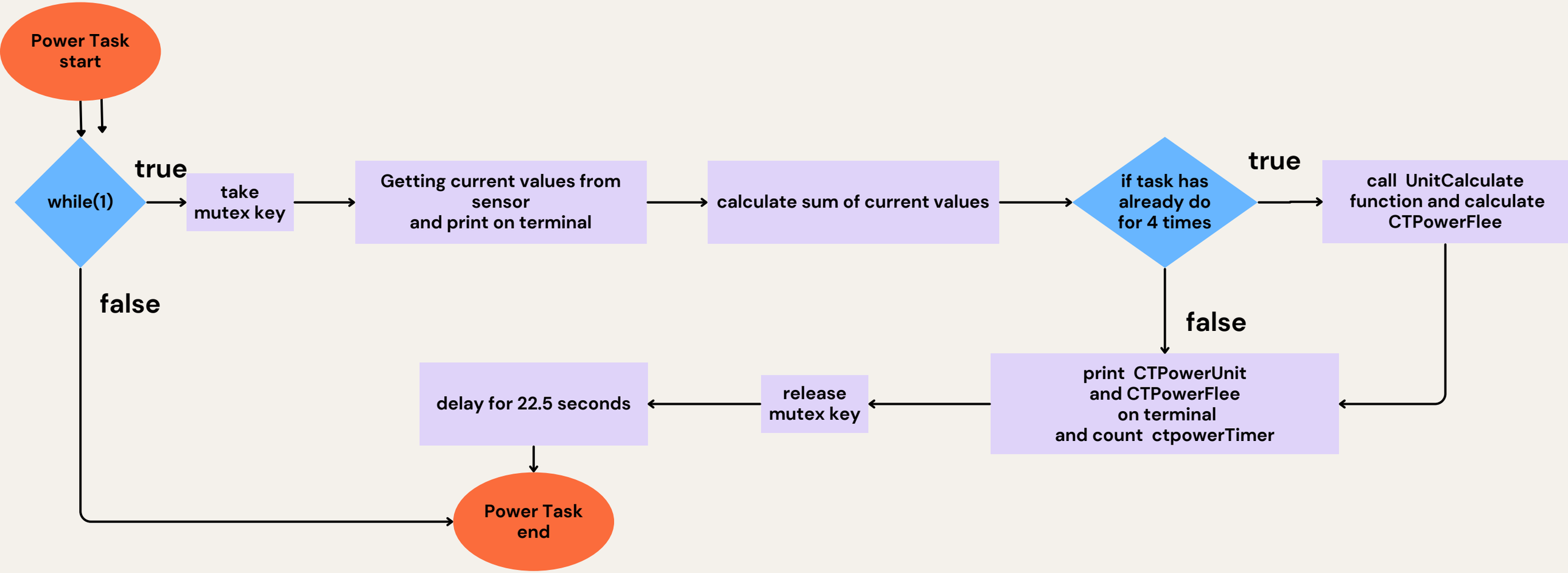
Flow Chart

Temp(Temperature)Task



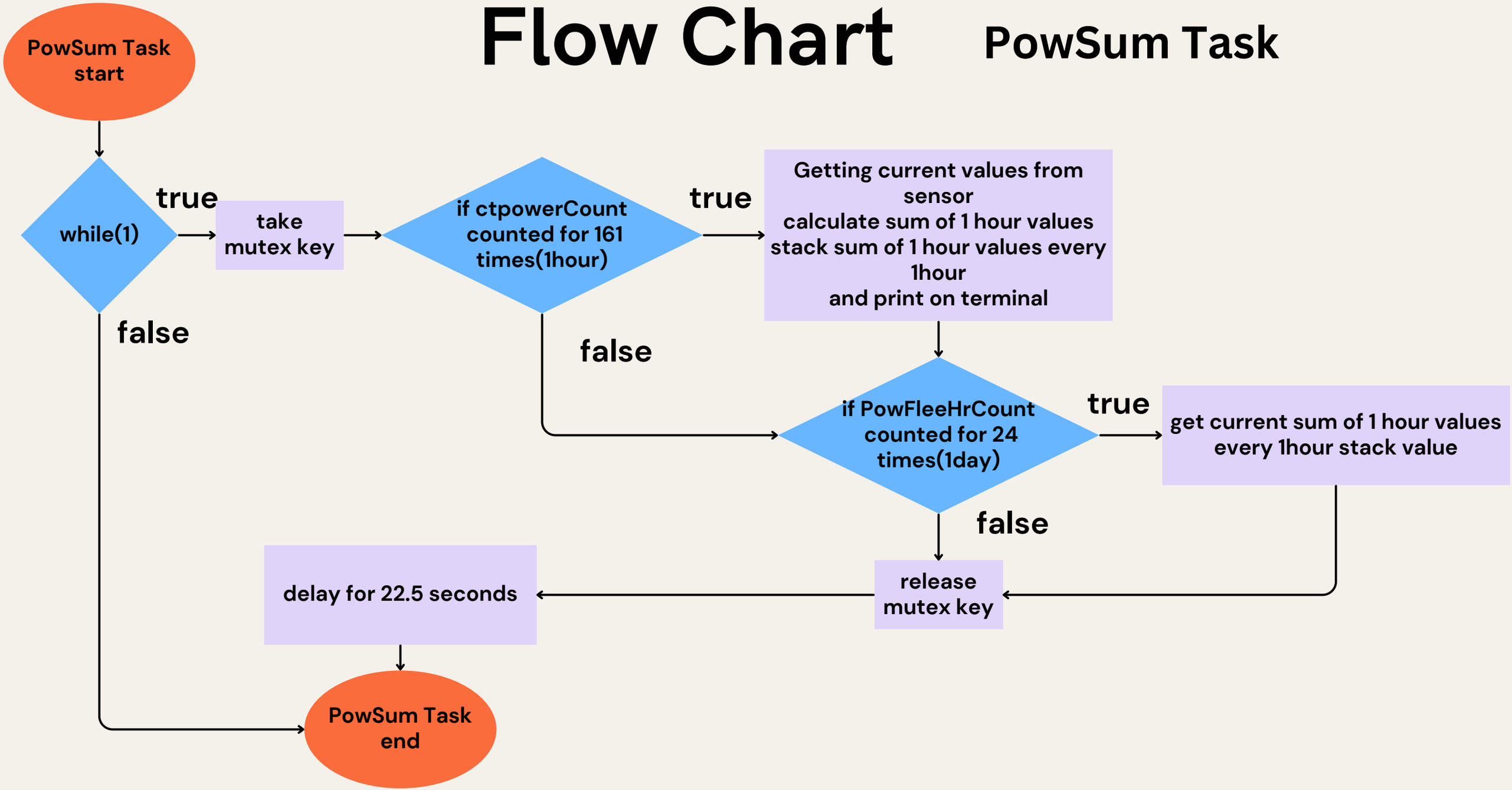
Flow Chart

Power Task



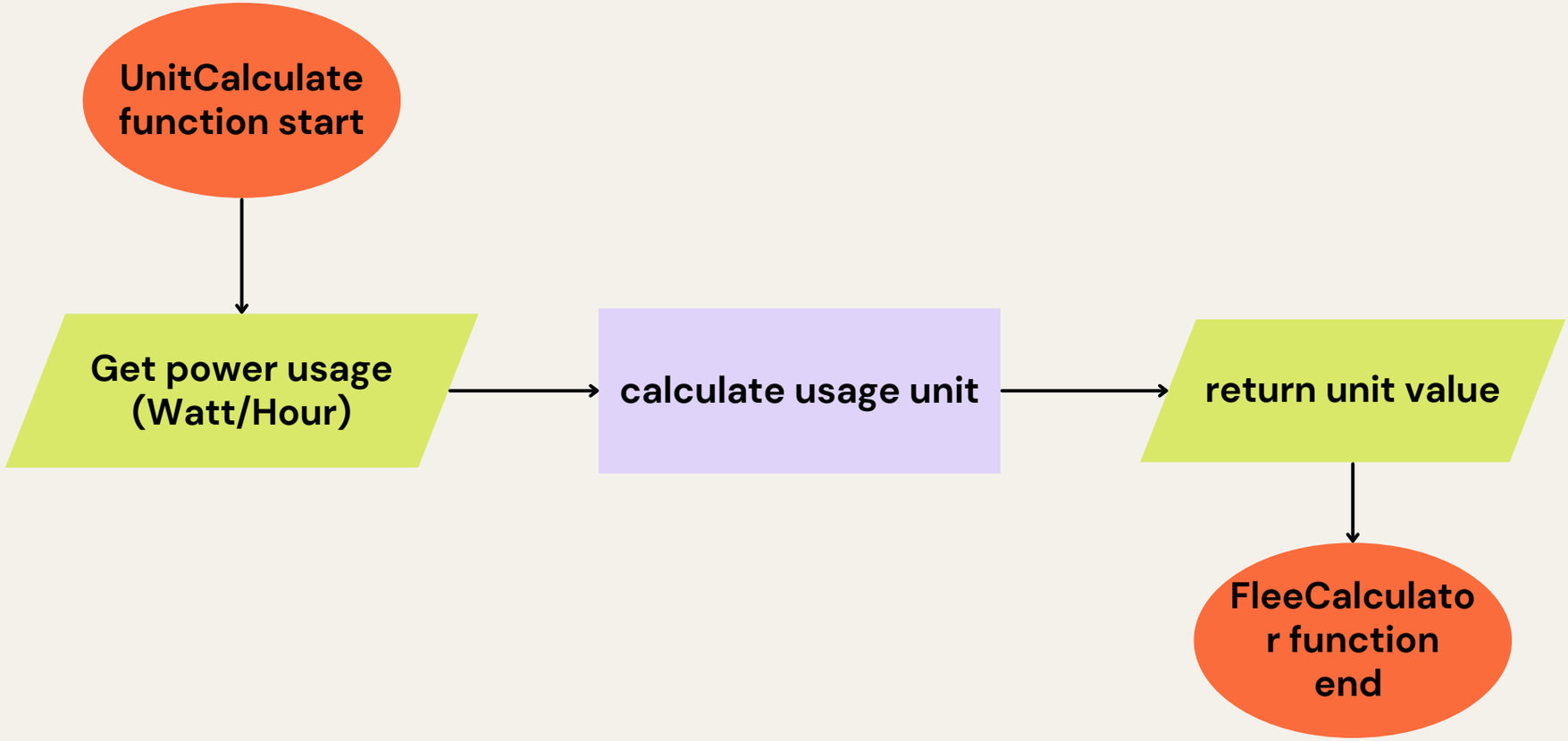
Flow Chart

PowSum Task



Flow Chart

UnitCalculate function



Flow Chart

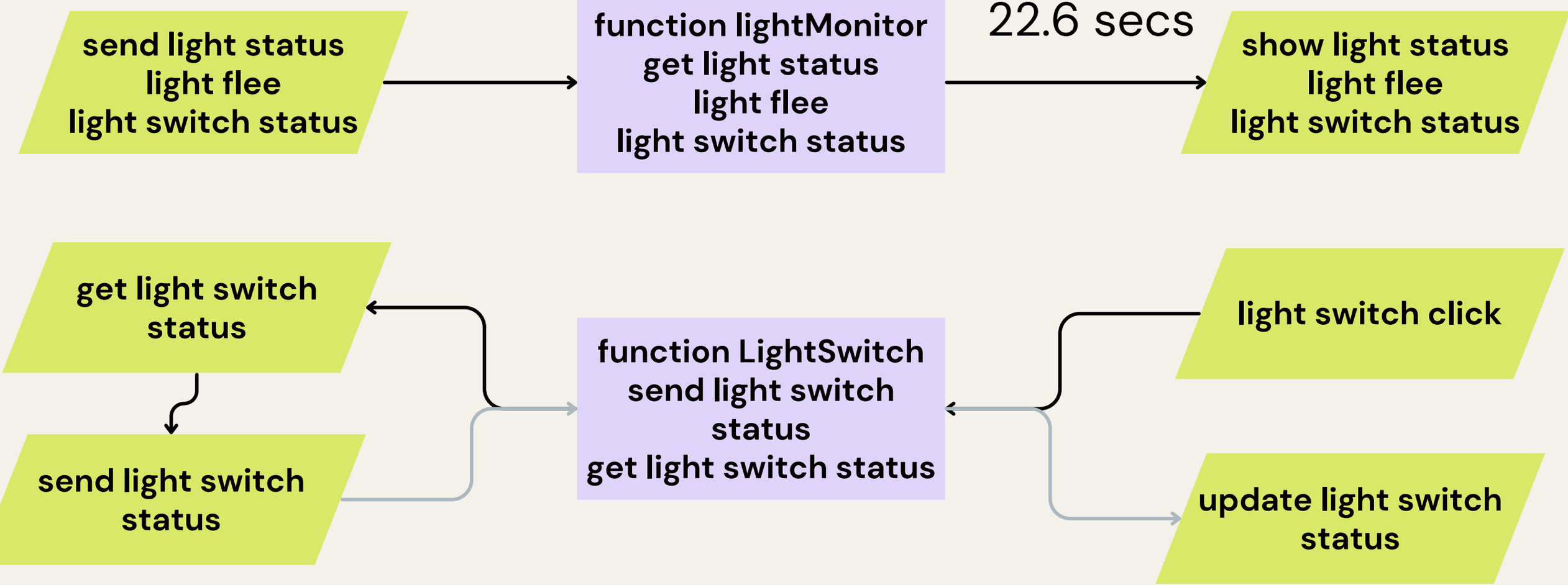
light data sending

ESP32

index.h JS

refresh every 22.6 secs

HTML



Flow Chart

air data sending

ESP32

send temperature
heatIndex
heatStatus
cooler switch status

index.h JS

function temperature
get temperature
heatIndex
heatStatus
cooler switch status

refresh every
22.6 secs

HTML

show temperature
heatIndex
heatStatus
cooler switch status

get cooler switch status

send cooler switch status

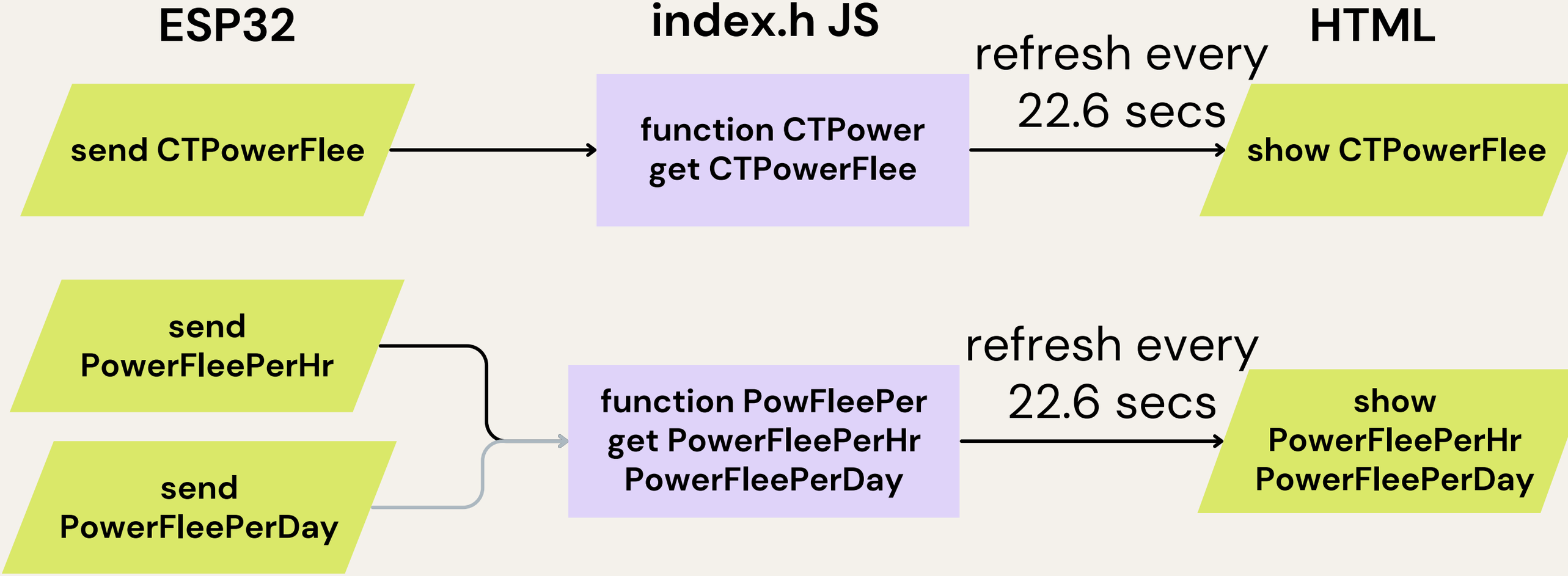
function AirconSwitch
send cooler switch status
get cooler switch status

cooler switch click

update cooler switch status

Flow Chart

ct power & power flee/hour data sending



Code

```
#include <Arduino.h>
#include <DHT.h>
#include <Adafruit_Sensor.h>
#include "EmonLib.h"
#include <WiFi.h>
#include <ESPAsyncWebServer.h>
#include "index.h"

#define DHTPIN 16 // Digital pin connected to the DHT sensor
#define DHTTYPE DHT22 // DHT 22 (AM2302), AM2321
DHT dht(DHTPIN, DHTTYPE);
int analogPin = 34, light = 0, lightTemp;
SemaphoreHandle_t xMutex;

int state = 0;

int LtaskPrio = 1, TtaskPrio = 1, lightCount = 0, airCount = 0, ctpowerCount = 0;
float lightUnit, powerFlee, lightPower = 46, CheckDelay = 22500;
float airUnit, usageUnit, airFlee, airPower = 782.81;
float CTPower, CTPowerUnit, CTPowerFlee;
float PowFleePerHr, PowFleePerHr_Temp;
String lightStatus, heatStatus;

float h, t, hic;

EnergyMonitor emon1;
EnergyMonitor emon2;

String airSliderValue = "OFF";
String lightSliderValue = "OFF";

const char* PARAM_INPUT = "value";

const char* WIFI_NAME = "Homewifi_2.4G";
const char* WIFI_PASSWORD = "No0955653261";
AsyncWebServer server(80);
```

ส่วนของ การนำเข้าlibrary และ การประกาศตัวประเบื้องต้น

ส่วนของ UnitCalculate function

```
38  float UnitCalculate(float power){
39      usageUnit = power*(CheckDelay/3600000)/1000;
40      return usageUnit;
41  }
```

Code

ส่วนของ setup

กำหนด PIN และ เชื่อมต่อ WIFI

```
184 void setup(){
185   Serial.begin(9600);
186   pinMode(analogPin, INPUT);
187   emon1.current(35, 111.1);
188   emon2.current(39, 111.1);
189   dht.begin();
190 
191   // Connect to Wi-Fi
192   WiFi.begin(WIFI_NAME, WIFI_PASSWORD);
193   while (WiFi.status() != WL_CONNECTED) {
194     delay(1000);
195     Serial.println("Connecting to WiFi...");
196   }
197   Serial.println("Connected to WiFi");
198 
199   // Print the ESP32's IP address
200   Serial.print("ESP32 Web Server's IP address: ");
201   Serial.println(WiFi.localIP());
```

สร้าง Mutex และ Task

```
312   xMutex = xSemaphoreCreateMutex();
313 
314   xTaskCreate(lightTask, "lightTask", 2048, NULL, LtaskPrio, NULL);
315   xTaskCreate(TempTask, "TempTask", 4096, NULL, TtaskPrio, NULL);
316   xTaskCreate(Power, "Power", 8192, NULL, TtaskPrio, NULL);
317   xTaskCreate(PowPerHr, "PowPerHr", 4096, NULL, LtaskPrio, NULL);
318 
319   }
```

สั่งสร้างหน้าHTML

```
203   // Serve the HTML page from the file
204   server.on("/", HTTP_GET, [](AsyncWebServerRequest* request) {
205     Serial.println("ESP32 Web Server: New request received:");  // for debugging
206     Serial.println("GET /");                                    // for debugging
207 
208     request->send(200, "text/html", webpage);
209   });
```

เปิด webserver

```
309   // Start the server
310   server.begin();
```

Code ส่วนของ lightTask

```
43  void lightTask(void *param){
44    while(1){
45      xSemaphoreTake( xMutex,portMAX_DELAY);
46      light = analogRead(analogPin);
47      Serial.println(light);
48      if(light < 3800){
49        lightStatus = "Bright!";
50        Serial.println(lightStatus);
51      }
52      else{
53        if(lightSliderValue == "OFF")
54        lightStatus = "Dark! turn on the light?";
55        Serial.println(lightStatus);
56      }
57
58      if(lightSliderValue == "ON"){
59        Serial.printf("\n\t\tlight is %s", lightSliderValue);
60        if(lightCount > 0){
61          lightUnit += UnitCalculate(lightPower);
62          powerFlee = lightUnit * 3.96;
63        }
64        Serial.printf("%.4f\n",lightUnit);
65        Serial.printf("%.4f\n",powerFlee);
66        lightCount++;
67      }
68      else if(lightSliderValue == "OFF"){
69        Serial.printf("\n\t\tlight is %s", lightSliderValue);
70        lightCount = 0;
71      }
72
73      xSemaphoreGive( xMutex ); //release key(xMutex)
74      vTaskDelay(pdMS_TO_TICKS(CheckDelay));
75    }
76  }
```

ส่วนของ lightTask ใน setup() เพื่อส่งข้อมูลไปหน้าเว็บ

```
211  // Define a route to get the light data
212  server.on("/light", HTTP_GET, [](AsyncWebServerRequest* request) {
213    Serial.println("ESP32 Web Server: New request received:");  // for debugging
214    Serial.println("GET /light");                               // for debugging
215    request->send(200, "text/plain", lightStatus);
216  });
217
218  server.on("/LightSwitchStatus", HTTP_GET, [](AsyncWebServerRequest* request) {
219    Serial.println("ESP32 Web Server: New request received:");  // for debugging
220    Serial.println("GET /LightSwitchStatus");                   // for debugging
221    request->send(200, "text/plain", lightSliderValue);
222  });
223
224  server.on("/lightFlee", HTTP_GET, [](AsyncWebServerRequest* request) {
225    Serial.println("ESP32 Web Server: New request received:");  // for debugging
226    Serial.println("GET /lightFlee");                           // for debugging
227    String lightFlee = String(powerFlee, 4);
228    request->send(200, "text/plain", lightFlee);
229  });
```

Code

ส่วนของ lightTask

ส่วนของ lightTask ใน setup() เพื่อรับข้อมูลจากหน้าเว็บ

```
295   server.on("/lightSlider", HTTP_GET, [] (AsyncWebServerRequest *request) {
296     String LightInputMessage;
297     // GET input1 value on <ESP_IP>/slider?value=<inputMessage>
298     if (request->hasParam(PARAM_INPUT)) {
299       LightInputMessage = request->getParam(PARAM_INPUT)->value();
300       lightSliderValue = LightInputMessage;
301     }
302     else {
303       LightInputMessage = "No message sent";
304     }
305     Serial.println(LightInputMessage);
306     request->send(200, "text/plain", "OK");
307   });
```

ส่วนของ lightTask ใน index.h js เพื่อรับข้อมูลจากESP32

```
128   <script>
129       function lightMonitor() {
130           fetch("/light")
131               .then(response => response.text())
132               .then(data => {
133                   document.getElementById("light").textContent = data;
134               });
135
136           fetch("/lightFlee")
137               .then(response => response.text())
138               .then(data => {
139                   document.getElementById("lightFlee").textContent = data;
140               });
141
142           fetch("/LightSwitchStatus")
143               .then(response => response.text())
144               .then(data => {
145                   document.getElementById("LightSwitchStatus").textContent = data;
146               });
147       }
148
149       lightMonitor();
150       setInterval(lightMonitor, 22600); // Update temperature every 4 seconds
```

Code

ส่วนของ lightTask

ส่วนของ lightTask ใน index.h js เพื่อส่งข้อมูลไปESP32

```
153     function LightSwitch() {
154         var LightCheckBox = document.getElementById("LightSwitchCheck");
155         var Lightxhr = new XMLHttpRequest();
156         if (LightCheckBox.checked == true){
157             Lightxhr.open("GET", "/lightSlider?value=ON", true);
158         } else {
159             Lightxhr.open("GET", "/lightSlider?value=OFF", true);
160         }
161         Lightxhr.send();
162 
163         fetch("/LightSwitchStatus")
164             .then(response => response.text())
165             .then(data => {
166                 document.getElementById("LightSwitchStatus").textContent = data;
167             });
168     }
```

ส่วนของการแสดงผล lightTask ใน index.h

```
89  <body>
90      <h1>Home power usage monitor</h1>
91      <i class="fa fa-lightbulb-o" style="font-size:48px; color:#ffca04;"></i>
92      <h2>
93          Your room is now <span style="color: red;"><span id="light">Loading...</span></span>
94      </h2>
95      <h2>
96          Light switch is turn <span style="color: red;"><span id="LightSwitchStatus">Loading...</span></span>
97          <label class="switch">
98              <input type="checkbox" id="LightSwitchCheck" onclick="LightSwitch()">
99              <span class="slider round"></span>
100         </label>
101     </h2>
102     <h2>
103         Current Power Flee: <span style="color: red;"><span id="lightFlee">Loading...</span></span>THB
104     </h2>
105     <br>
```

Code
ส่วนของ TempTask

```
78  void TempTask(void *param){
79    while(1){
80      xSemaphoreTake( xMutex,portMAX_DELAY);
81      // Reading temperature or humidity takes about 250 milliseconds!
82      // Sensor readings may also be up to 2 seconds 'old' (its a very slow sensor)
83      h = dht.readHumidity();
84      // Read temperature as Celsius (the default)
85      t = dht.readTemperature();
86
87      // Check if any reads failed and exit early (to try again).
88      if (isnan(h) || isnan(t)) {
89      Serial.println(F("Failed to read from DHT sensor!"));
90      return;
91      }
92
93      // Compute heat index in Celsius (isFahreheit = false)
94      hic = dht.computeHeatIndex(t, h, false);
95
96      Serial.print(F("\t\t\tHumidity: "));
97      Serial.print(h);
98      Serial.print(F("% Temperature: "));
99      Serial.print(t);
100     Serial.print(F(" C "));
101     Serial.print(F(" Heat index: "));
102     Serial.print(hic);
103     Serial.print(F(" C "));
104
105     if(hic >= 32){
106       if(airSliderValue == "OFF")
107       heatStatus = "so HOT! turn on the cooler?";
108       else heatStatus = " ";
109       Serial.println(heatStatus);
110     }
111     else if(hic < 17){
112       if(airSliderValue == "ON")
113       heatStatus = "so COLD! turn off the cooler?";
114       else heatStatus = " ";
115       Serial.println(heatStatus);
116     }
117     else{
118       heatStatus = "COOL!";
119       Serial.println(heatStatus);
```

```
99      Serial.print(t);
100     Serial.print(F(" C "));
101     Serial.print(F(" Heat index: "));
102     Serial.print(hic);
103     Serial.print(F(" C "));
104
105     if(hic >= 32){
106       if(airSliderValue == "OFF")
107       heatStatus = "so HOT! turn on the cooler?";
108       else heatStatus = " ";
109       Serial.println(heatStatus);
110     }
111     else if(hic < 17){
112       if(airSliderValue == "ON")
113       heatStatus = "so COLD! turn off the cooler?";
114       else heatStatus = " ";
115       Serial.println(heatStatus);
116     }
117     else{
118       heatStatus = "COOL!";
119       Serial.println(heatStatus);
120     }
121
122     if(airSliderValue == "ON"){
123       Serial.printf("\n\t\tair is %s", airSliderValue);
124       if(airCount > 0){
125         airUnit += UnitCalculate(airPower);
126         airFlee = airUnit * 3.96;
127       }
128       Serial.printf("AU:%.4f\n",airUnit);
129       Serial.printf("AF:%.4f\n",airFlee);
130       airCount++;
131     }
132     else if(airSliderValue == "OFF"){
133       Serial.printf("\n\t\tair is %s", airSliderValue);
134       airCount = 0;
135     }
136
137     xSemaphoreGive( xMutex ); //release key(xMutex)
138     vTaskDelay(pdMS_TO_TICKS(CheckDelay));
139   }
140 }
```

Code ส่วนของ TempTask

ส่วนของ TempTask ใน setup() เพื่อส่งข้อมูลไปหน้าเว็บ

```
235  server.on("/temperature", HTTP_GET, [](AsyncWebServerRequest* request) {
236    Serial.println("ESP32 Web Server: New request received:");  // for debugging
237    Serial.println("GET /temperature");                         // for debugging
238    float temperature = t;
239    String temperatureStr = String(temperature, 2);
240    request->send(200, "text/plain", temperatureStr);
241  });
242
243  server.on("/heatIndex", HTTP_GET, [](AsyncWebServerRequest* request) {
244    Serial.println("ESP32 Web Server: New request received:");  // for debugging
245    Serial.println("GET /heatIndex");                          // for debugging
246    String heatIndecStr = String(hic, 2);
247    request->send(200, "text/plain", heatIndecStr);
248  });
249
250  server.on("/heatStatus", HTTP_GET, [](AsyncWebServerRequest* request) {
251    Serial.println("ESP32 Web Server: New request received:");  // for debugging
252    Serial.println("GET /heatStatus");                         // for debugging
253    request->send(200, "text/plain", heatStatus);
254  });
255
256  server.on("/airStatus", HTTP_GET, [](AsyncWebServerRequest* request) {
257    Serial.println("ESP32 Web Server: New request received:");  // for debugging
258    Serial.println("GET /airStatus");                          // for debugging
259    request->send(200, "text/plain", airSliderValue);
260  });
261
262  server.on("/airFlee", HTTP_GET, [](AsyncWebServerRequest* request) {
263    Serial.println("ESP32 Web Server: New request received:");  // for debugging
264    Serial.println("GET /airFlee");                            // for debugging
265    float af = airFlee;
266    String AirFlee = String(af, 4);
267    request->send(200, "text/plain", AirFlee);
268  });
```

ส่วนของ TempTask ใน setup() เพื่อรับข้อมูลจากหน้าเว็บ

```
292  server.on("/airSlider", HTTP_GET, [] (AsyncWebServerRequest *request) {
293      String AirInputMessage;
294      // GET input1 value on <ESP_IP>/slider?value=<inputMessage>
295      if (request->hasParam(PARAM_INPUT)) {
296        AirInputMessage = request->getParam(PARAM_INPUT)->value();
297        airSliderValue = AirInputMessage;
298      }
299      else {
300        AirInputMessage = "No message sent";
301      }
302      Serial.println(AirInputMessage);
303      request->send(200, "text/plain", "OK");
304    });
```

Code

ส่วนของ TempTask

ส่วนของ TempTask ใน index.h js เพื่อรับข้อมูลจากESP32

```
function temperature() {
    fetch("/temperature")
        .then(response => response.text())
        .then(data => {
            document.getElementById("temperature").textContent = data;
        });

    fetch("/heatIndex")
        .then(response => response.text())
        .then(data => {
            document.getElementById("heatIndex").textContent = data;
        });

    fetch("/heatStatus")
        .then(response => response.text())
        .then(data => {
            document.getElementById("heatStatus").textContent = data;
        });

    fetch("/airFlee")
        .then(response => response.text())
        .then(data => {
            document.getElementById("airFlee").textContent = data;
        });

    fetch("/airStatus")
        .then(response => response.text())
        .then(data => {
            document.getElementById("airStatus").textContent = data;
        });
}
```

ส่วนของการแสดงผล TempTask ใน index.h

```
<i class="fa fa-thermometer-half" style="font-size:48px; color:#059e8a;"></i>
<h2>
    Temperature: <span style="color: red;"><span id="temperature">Loading...</span>&deg</span>
    Heat Index: <span style="color: red;"><span id="heatIndex">Loading...</span>&deg</span>
</h2>
<h2>Your room is <span style="color: red;"><span id="heatStatus">Loading...</span></span>
<h2>
    Air conditioner is <span style="color: red;"><span id="airStatus">Loading...</span></span>
    <label class="switch">
        <input type="checkbox" id="AirSwitchCheck" onclick="AirconSwitch()">
        <span class="slider round"></span>
    </label>
</h2>
<h2>
    Air Flee: <span style="color: red;"><span id="airFlee">Loading...</span></span>THB
</h2>
<br>
```

ส่วนของ TempTask ใน index.h js เพื่อส่งข้อมูลไปESP32

```
function AirconSwitch() {
    var AirCheckBox = document.getElementById("AirSwitchCheck");
    var Airxhr = new XMLHttpRequest();
    if (AirCheckBox.checked == true){
        Airxhr.open("GET", "/airSlider?value=ON", true);
    } else {
        Airxhr.open("GET", "/airSlider?value=OFF", true);
    }
    Airxhr.send();

    fetch("/airStatus")
        .then(response => response.text())
        .then(data => {
            document.getElementById("airStatus").textContent = data;
        });
}
```

Code

ส่วนของ Power Task

```
143  void Power(void *parameter){
144    while(1){
145      xSemaphoreTake( xMutex,portMAX_DELAY);
146      double Irms = emon1.calcIrms(1480);  // Calculate Irms only
147      Serial.print(Irms*230.0);         // Apparent power
148      Serial.print(" ");
149      Serial.println(Irms);          // Irms
150
151      double Irmss = emon2.calcIrms(1480);  // Calculate Irms only
152      Serial.print(Irmss*230.0);         // Apparent power
153      Serial.print(" ");
154      Serial.println(Irmss);           // Irms
155
156      CTPower = (Irms + Irmss)*230;
157
158      if(ctpowerCount > 3){
159        CTPowerUnit += UnitCalculate(CTPower);
160        CTPowerFlee = CTPowerUnit * 3.96;
161      }
162      Serial.printf("PU:%.4f\n",CTPowerUnit);
163      Serial.printf("PF:%.4f\n",CTPowerFlee);
164      ctpowerCount++;
165
166      Serial.println(uxTaskGetStackHighWaterMark(NULL));
167      xSemaphoreGive( xMutex ); //release key(xMutex)
168      vTaskDelay(pdMS_TO_TICKS(CheckDelay));
169    }
170  }
```

ส่วนของ PowSum Task

```
172  void PowSum(void *param){
173    while(1){
174      xSemaphoreTake( xMutex,portMAX_DELAY);
175      if(ctpowerCount % 161 == 0){
176        PowFleePerHr = powerFlee + airFlee + CTPowerFlee;
177        PowFleePerHr_Temp += PowFleePerHr;
178        PowFleeHrCount++;
179      }
180      if(PowFleeHrCount% 24 == 0){
181        PowFleePerDay = PowFleePerHr_Temp;
182      }
183      xSemaphoreGive( xMutex ); //release key(xMutex)
184      vTaskDelay(pdMS_TO_TICKS(CheckDelay));
185    }
186  }
```

Code

ส่วนของ Power Task และ PowSum Task

```
270  server.on("/CTPowerFlee", HTTP_GET, [](AsyncWebServerRequest* request) {
271    Serial.println("ESP32 Web Server: New request received:");  // for debugging
272    Serial.println("GET /CTPowerFlee");                         // for debugging
273    float CTPF = CTPowerFlee;
274    String CTPFlee = String(CTPF, 4);
275    request->send(200, "text/plain", CTPFlee);
276  });
```

ส่วนของ Power Task ใน setup() เพื่อส่งข้อมูลไปหน้าเว็บ

ส่วนของ PowerSum Task ใน setup() เพื่อส่งข้อมูลไปหน้าเว็บ

```
278  server.on("/PowerFleePerHr", HTTP_GET, [](AsyncWebServerRequest* request) {
279    Serial.println("ESP32 Web Server: New request received:");  // for debugging
280    Serial.println("GET /PowerFleePerHr");                      // for debugging
281    String PFleePerHr = String(PowFleePerHr, 4);
282    request->send(200, "text/plain", PFleePerHr);
283  });
284
285  server.on("/PowerFleePerDay", HTTP_GET, [](AsyncWebServerRequest* request) {
286    Serial.println("ESP32 Web Server: New request received:");  // for debugging
287    Serial.println("GET /PowerFleePerHr");                      // for debugging
288    String PFleePerDay = String(PowFleePerDay, 4);
289    request->send(200, "text/plain", PFleePerDay);
290  });
```

Code
ส่วนของ TempTask

ส่วนของการแสดงผล Power Task ใน index.h

```
123     <i class="fa fa-flash" style="font-size:48px;color:#00F0FF;"></i>
124     <h2>Power usage Flee: <span style="color: red;"><span id="CTPowerFlee">Loading...</span></span>THB</h2>
```

ส่วนของการแสดงผล PowerSum Task ใน index.h

```
126     <h2>Power Flee/Hour: <span style="color: red;"><span id="PowerFleePerHr">Loading...</span></span>THB</h2>
127     <h2>Power Flee/Day: <span style="color: red;"><span id="PowerFleePerDay">Loading...</span></span>THB</h2>
```

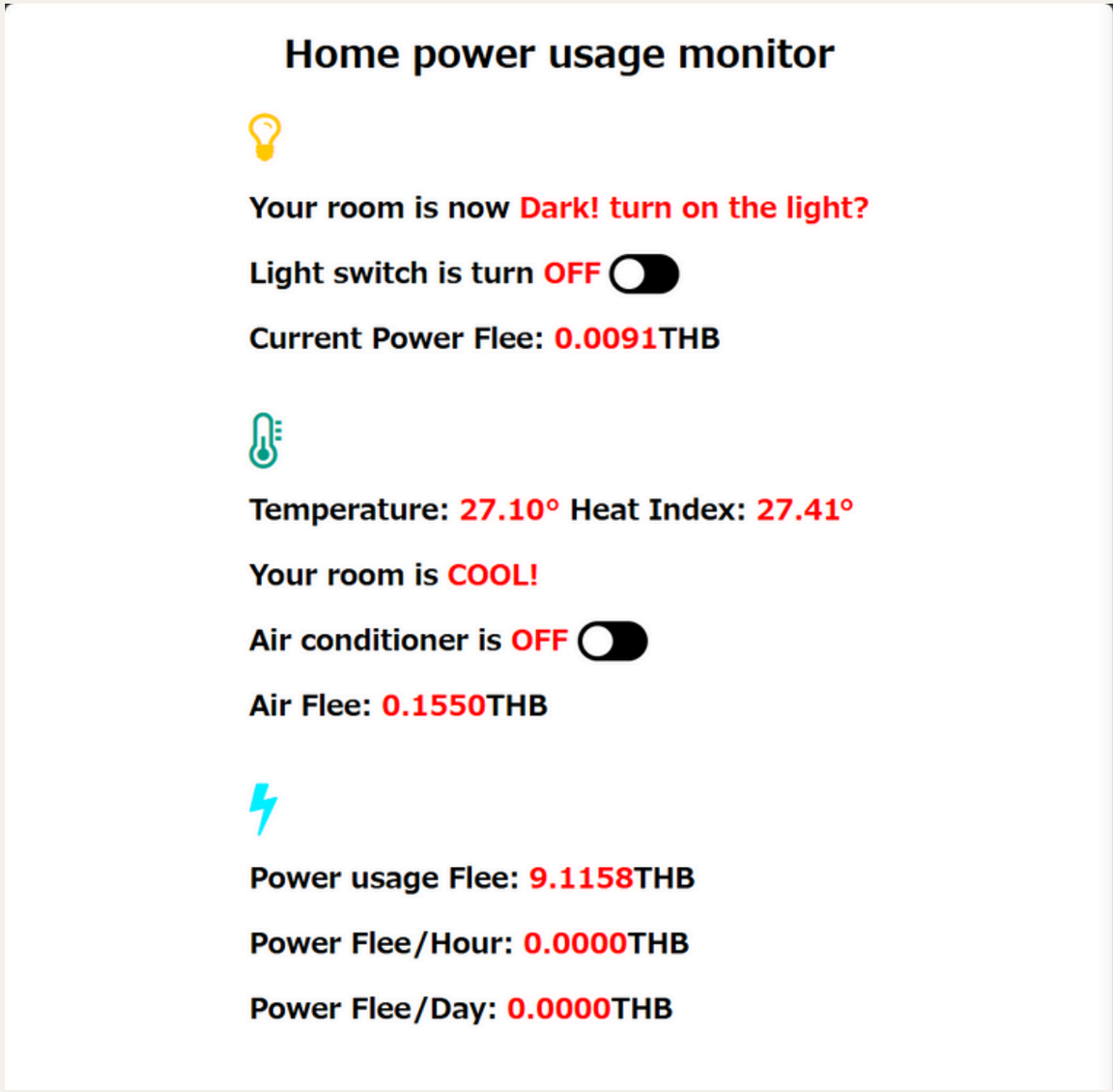
ส่วนของ Power Task ใน index.h js เพื่อรับข้อมูลจากESP32

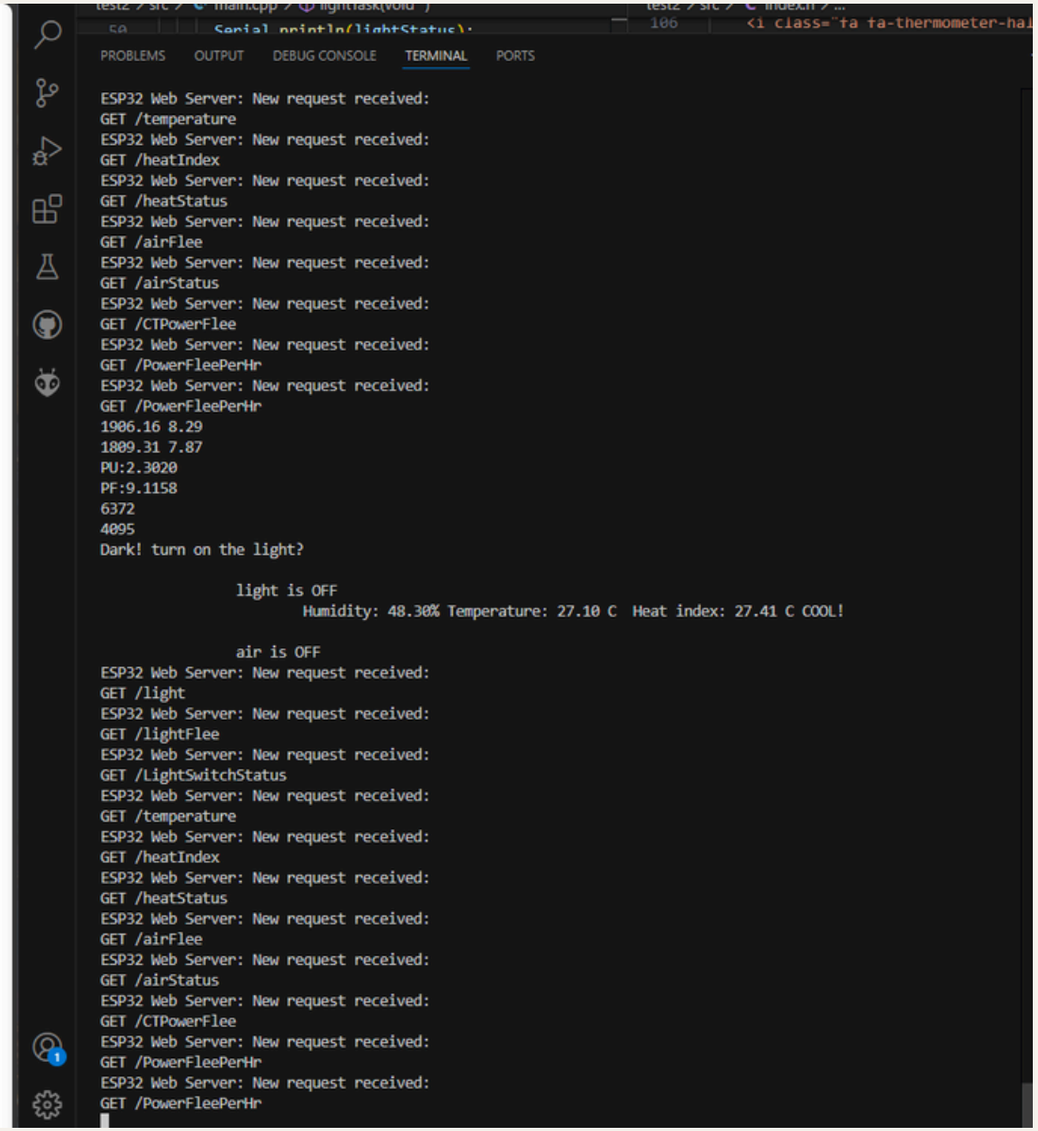
```
222     function CTPower() {
223         fetch("/CTPowerFlee")
224             .then(response => response.text())
225             .then(data => {
226                 document.getElementById("CTPowerFlee").textContent = data;
227             });
228     }
```

ส่วนของ PowerSum Task ใน index.h js เพื่อรับข้อมูลจากESP32

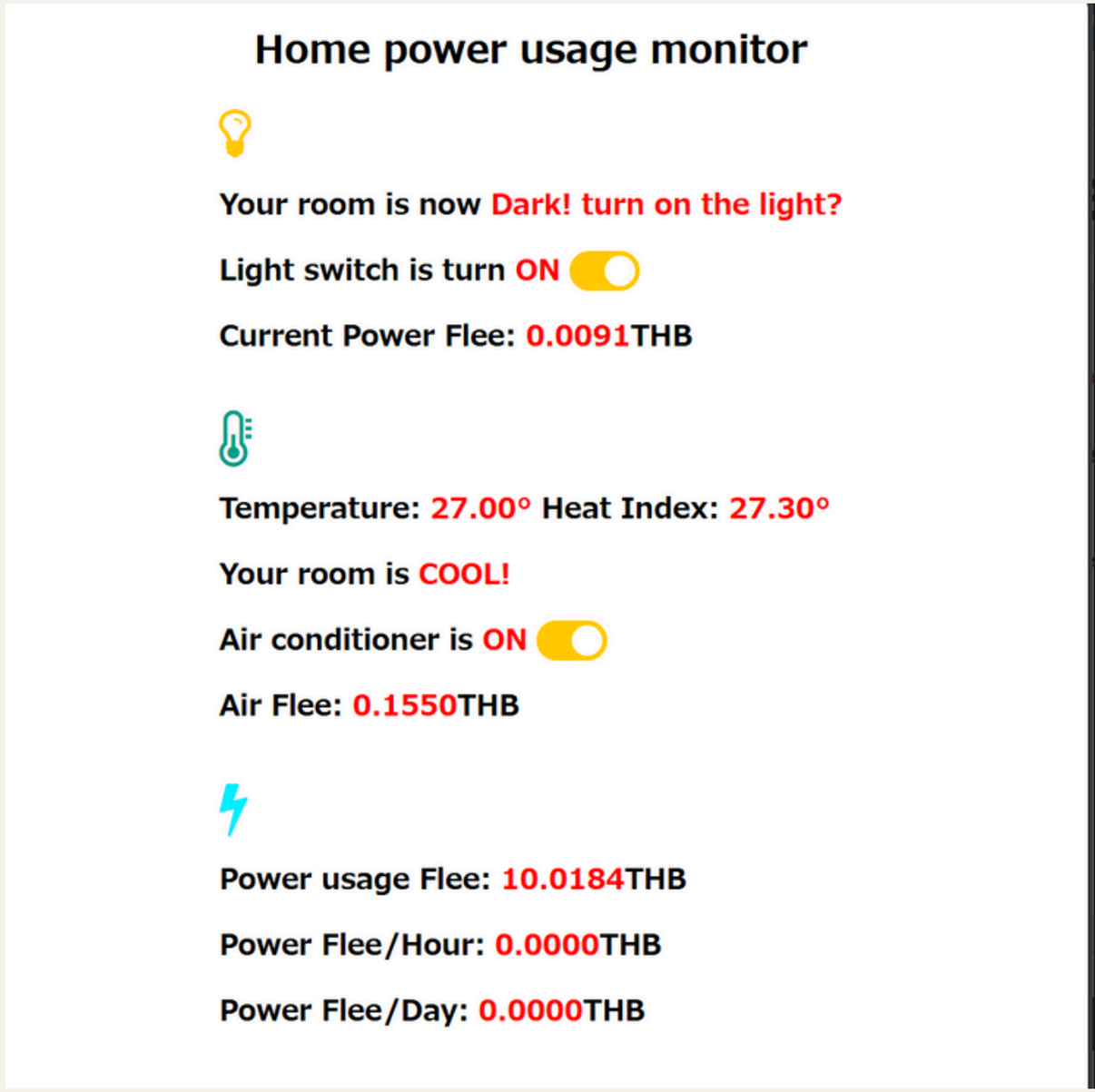
```
233     function PowFleePer() {
234         fetch("/PowerFleePerHr")
235             .then(response => response.text())
236             .then(data => {
237                 document.getElementById("PowerFleePerHr").textContent = data;
238             });
239
240         fetch("/PowerFleePerDay")
241             .then(response => response.text())
242             .then(data => {
243                 document.getElementById("PowerFleePerDay").textContent = data;
244             });
245     }
246
247     PowFleePer();
248     setInterval(PowFleePer, 22600); // Update temperature every 4 seconds
249 </script>
```

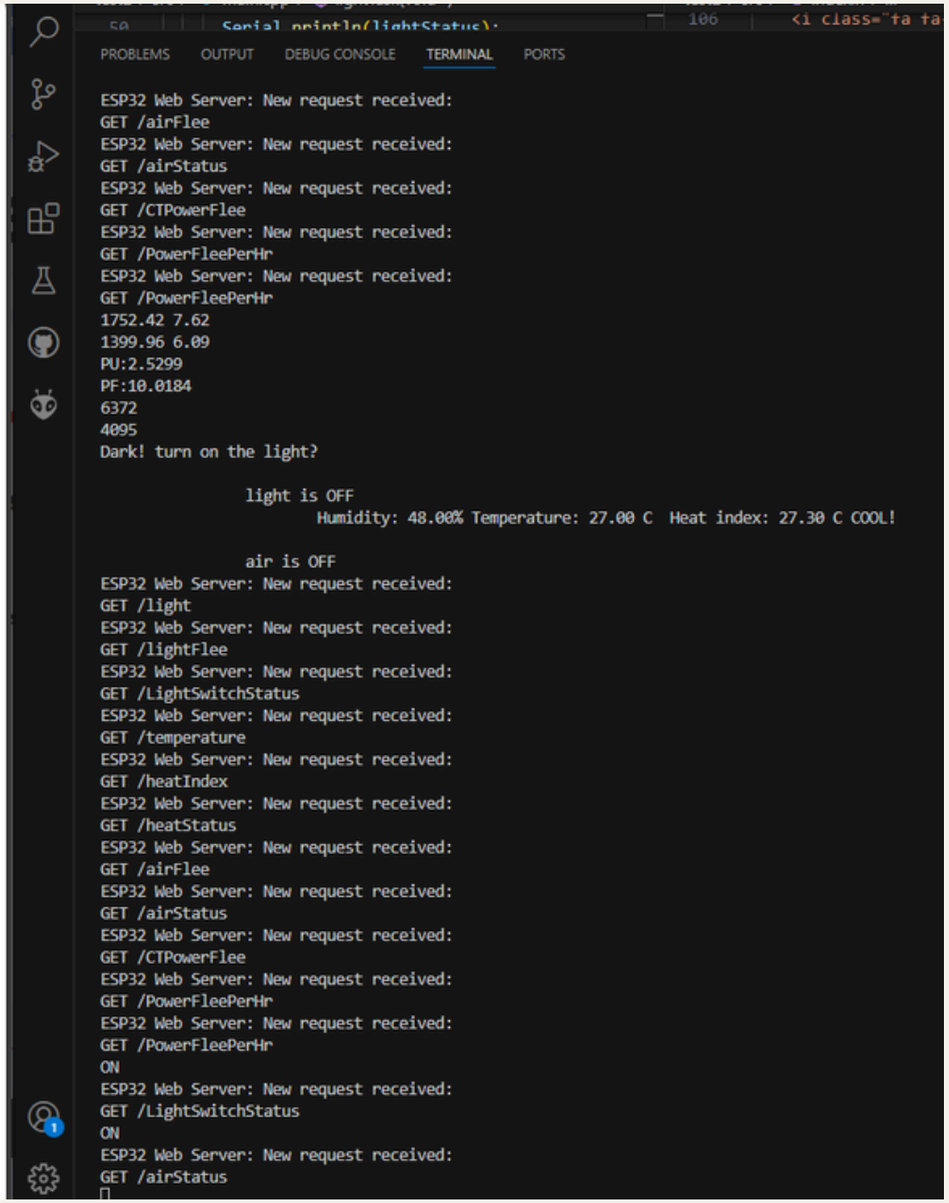
Test & Result


Home power usage monitor
Your room is now Dark! turn on the light?
Light switch is turn OFF
Current Power Flee: 0.0091THB
Temperature: 27.10° Heat Index: 27.41°
Your room is COOL!
Air conditioner is OFF
Air Flee: 0.1550THB
Power usage Flee: 9.1158THB
Power Flee/Hour: 0.0000THB
Power Flee/Day: 0.0000THB

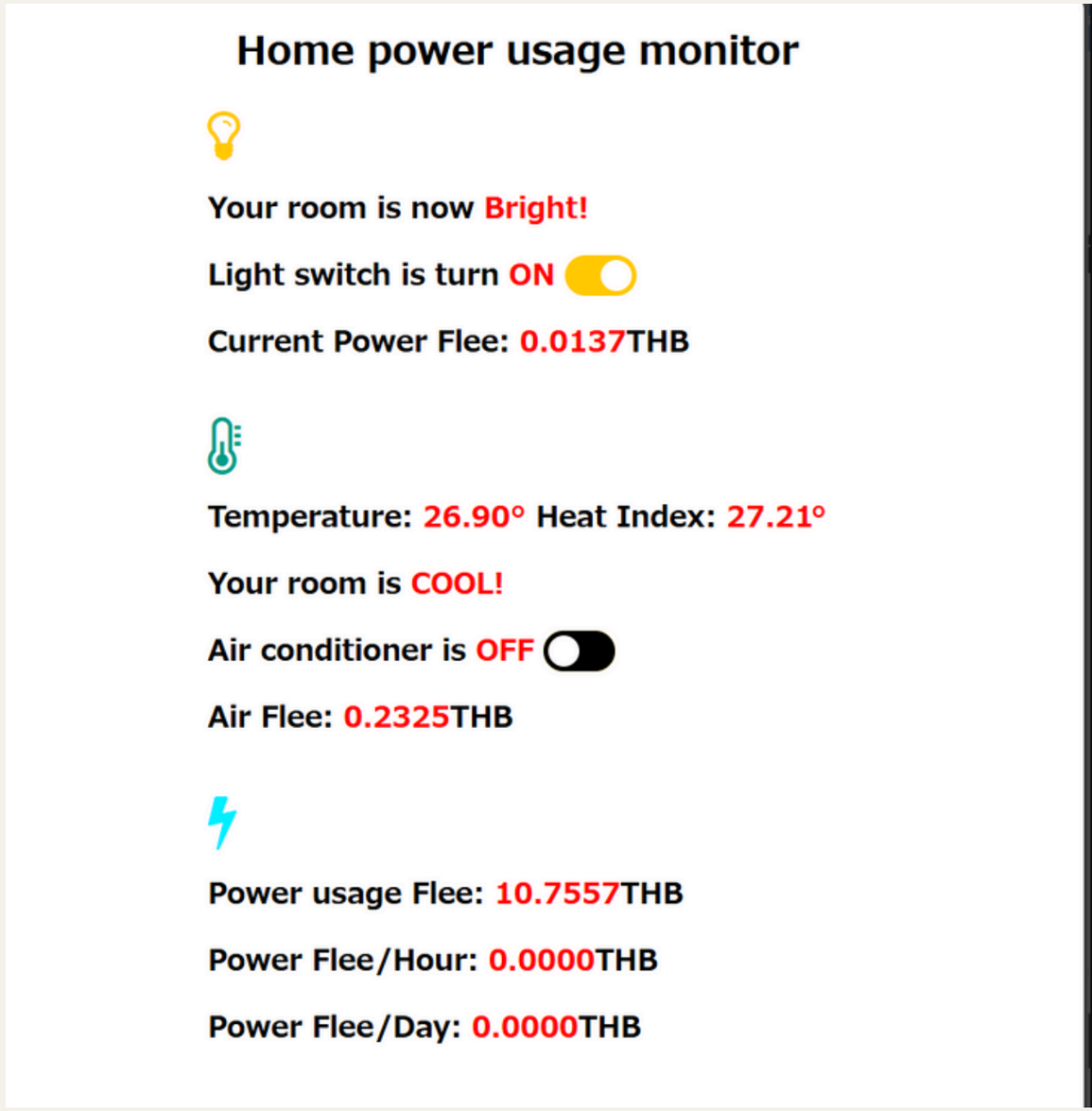

PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
ESP32 Web Server: New request received:
GET /temperature
ESP32 Web Server: New request received:
GET /heatIndex
ESP32 Web Server: New request received:
GET /heatStatus
ESP32 Web Server: New request received:
GET /airFlee
ESP32 Web Server: New request received:
GET /airStatus
ESP32 Web Server: New request received:
GET /CTPowerFlee
ESP32 Web Server: New request received:
GET /PowerFleePerHr
ESP32 Web Server: New request received:
GET /PowerFleePerHr
1906.16 8.29
1809.31 7.87
PU:2.3020
PF:9.1158
6372
4095
Dark! turn on the light?
light is OFF
Humidity: 48.30% Temperature: 27.10 C Heat index: 27.41 C COOL!
air is OFF
ESP32 Web Server: New request received:
GET /light
ESP32 Web Server: New request received:
GET /lightFlee
ESP32 Web Server: New request received:
GET /LightSwitchStatus
ESP32 Web Server: New request received:
GET /temperature
ESP32 Web Server: New request received:
GET /heatIndex
ESP32 Web Server: New request received:
GET /heatStatus
ESP32 Web Server: New request received:
GET /airFlee
ESP32 Web Server: New request received:
GET /airStatus
ESP32 Web Server: New request received:
GET /CTPowerFlee
ESP32 Web Server: New request received:
GET /PowerFleePerHr
ESP32 Web Server: New request received:
GET /PowerFleePerHr

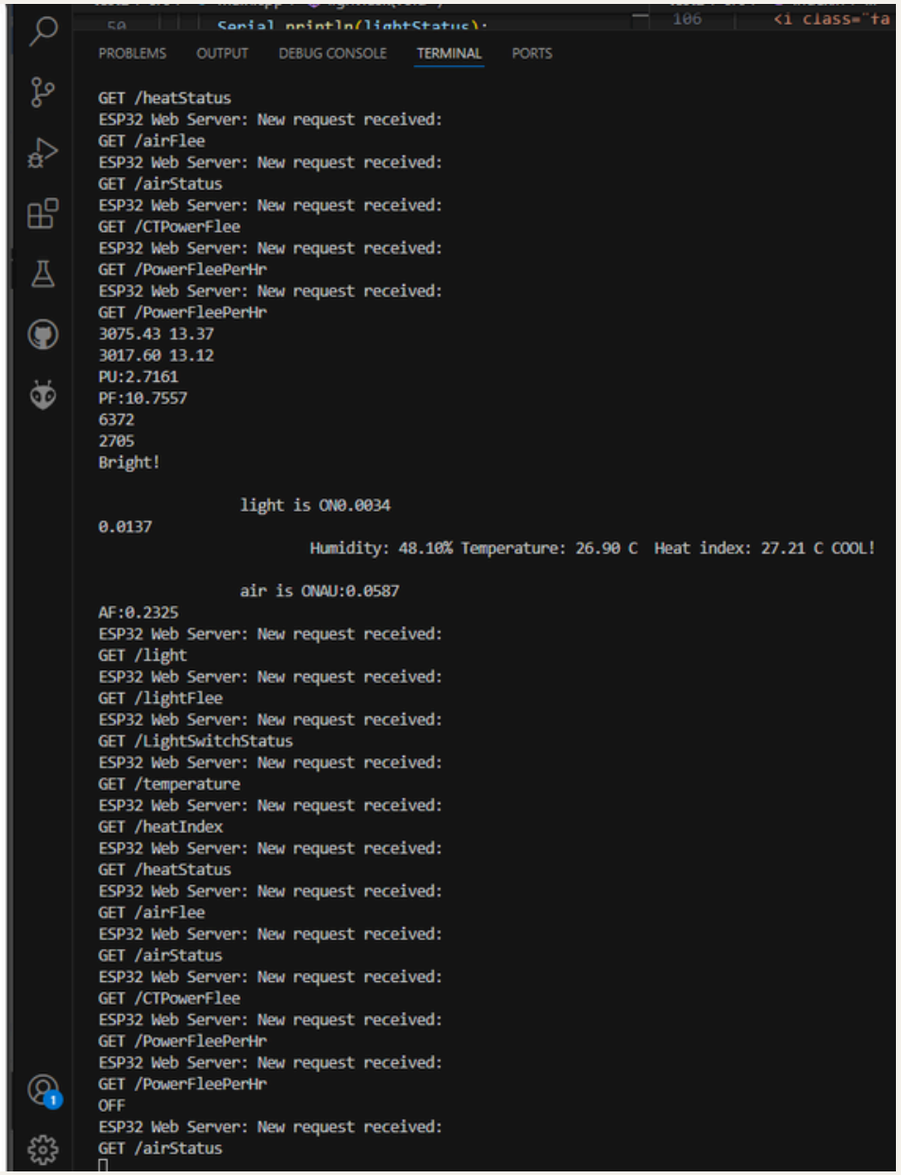
Test & Result


Home power usage monitor
Your room is now Dark! turn on the light?
Light switch is turn ON
Current Power Flee: 0.0091THB
Temperature: 27.00° Heat Index: 27.30°
Your room is COOL!
Air conditioner is ON
Air Flee: 0.1550THB
Power usage Flee: 10.0184THB
Power Flee/Hour: 0.0000THB
Power Flee/Day: 0.0000THB

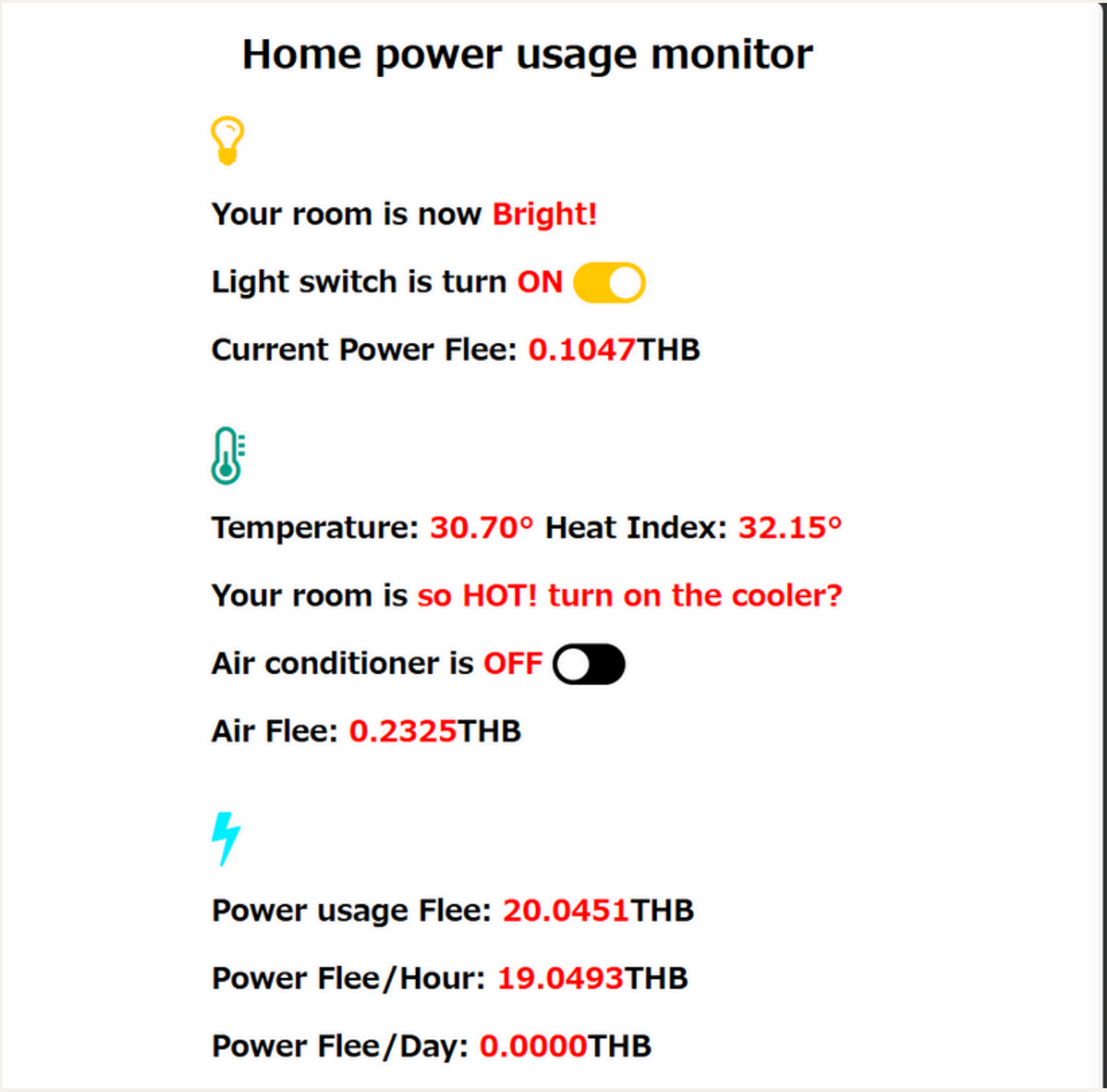

PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
ESP32 Web Server: New request received:
GET /airFlee
ESP32 Web Server: New request received:
GET /airStatus
ESP32 Web Server: New request received:
GET /CTPowerFlee
ESP32 Web Server: New request received:
GET /PowerFleePerHr
ESP32 Web Server: New request received:
GET /PowerFleePerHr
1752.42 7.62
1399.96 6.09
PU:2.5299
PF:10.0184
6372
4095
Dark! turn on the light?
light is OFF
Humidity: 48.00% Temperature: 27.00 C Heat index: 27.30 C COOL!
air is OFF
ESP32 Web Server: New request received:
GET /light
ESP32 Web Server: New request received:
GET /lightFlee
ESP32 Web Server: New request received:
GET /LightSwitchStatus
ESP32 Web Server: New request received:
GET /temperature
ESP32 Web Server: New request received:
GET /heatIndex
ESP32 Web Server: New request received:
GET /heatStatus
ESP32 Web Server: New request received:
GET /airFlee
ESP32 Web Server: New request received:
GET /airStatus
ESP32 Web Server: New request received:
GET /CTPowerFlee
ESP32 Web Server: New request received:
GET /PowerFleePerHr
ESP32 Web Server: New request received:
GET /PowerFleePerHr
ON
ESP32 Web Server: New request received:
GET /LightSwitchStatus
ON
ESP32 Web Server: New request received:
GET /airStatus

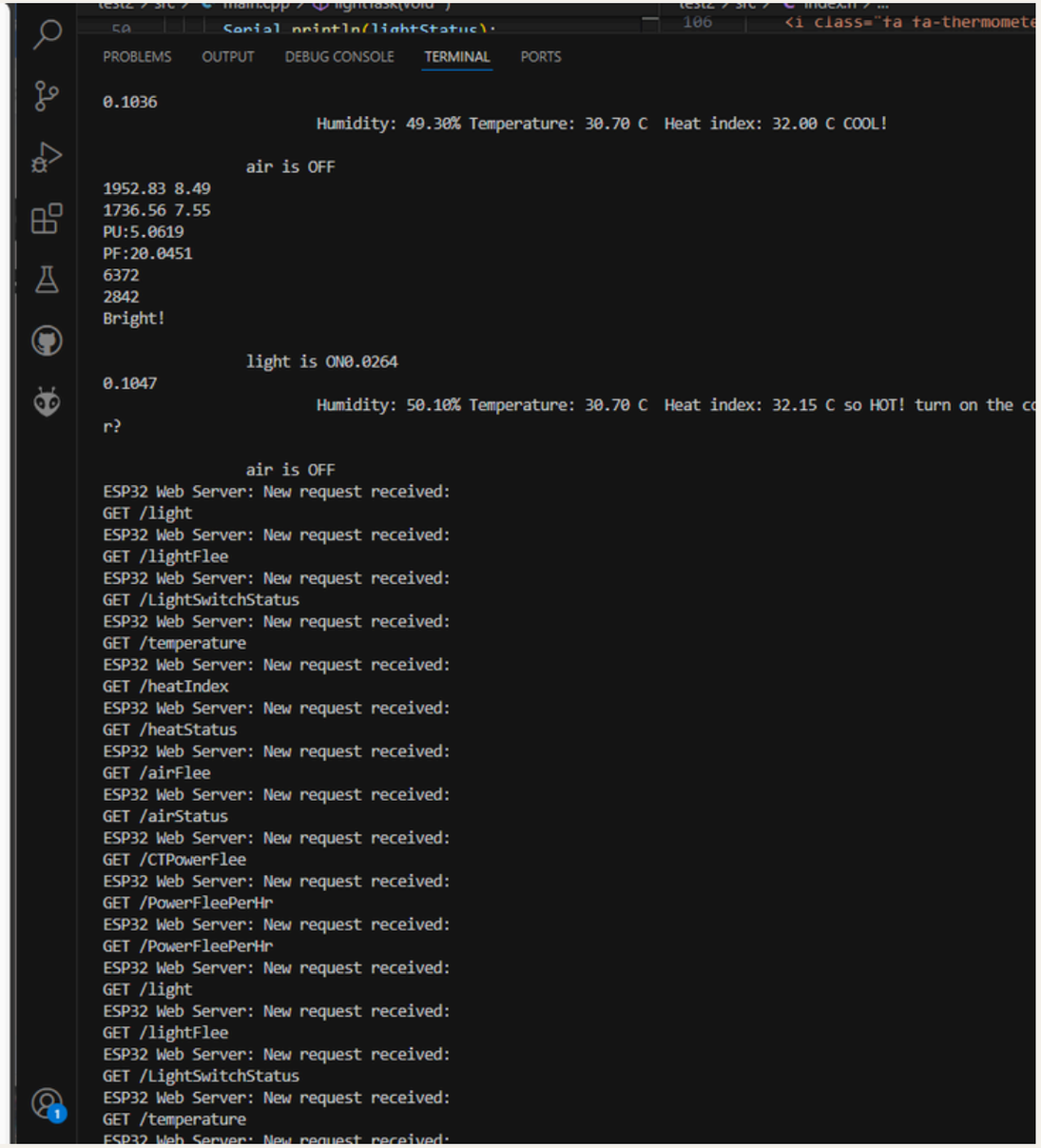
Test & Result


Home power usage monitor
Your room is now Bright!
Light switch is turn ON
Current Power Flee: 0.0137THB
Temperature: 26.90° Heat Index: 27.21°
Your room is COOL!
Air conditioner is OFF
Air Flee: 0.2325THB
Power usage Flee: 10.7557THB
Power Flee/Hour: 0.0000THB
Power Flee/Day: 0.0000THB

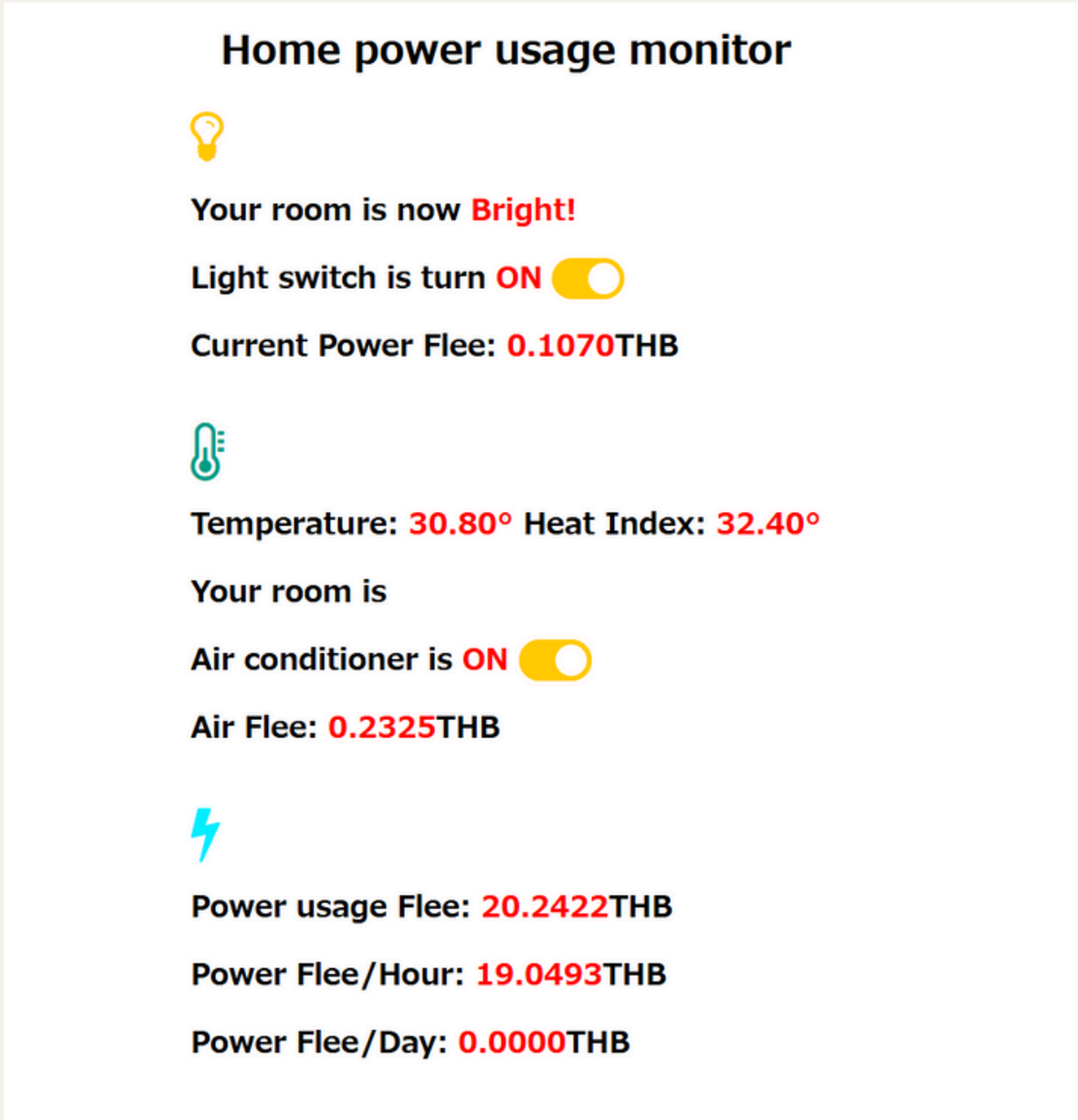

PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
GET /heatStatus
ESP32 Web Server: New request received:
GET /airFlee
ESP32 Web Server: New request received:
GET /airStatus
ESP32 Web Server: New request received:
GET /CTPowerFlee
ESP32 Web Server: New request received:
GET /PowerFleePerHr
ESP32 Web Server: New request received:
GET /PowerFleePerHr
3075.43 13.37
3017.60 13.12
PU:2.7161
PF:10.7557
6372
2705
Bright!
light is ON0.0034
0.0137
Humidity: 48.10% Temperature: 26.90 C Heat index: 27.21 C COOL!
air is ONAU:0.0587
AF:0.2325
ESP32 Web Server: New request received:
GET /light
ESP32 Web Server: New request received:
GET /lightFlee
ESP32 Web Server: New request received:
GET /LightSwitchStatus
ESP32 Web Server: New request received:
GET /temperature
ESP32 Web Server: New request received:
GET /heatIndex
ESP32 Web Server: New request received:
GET /heatStatus
ESP32 Web Server: New request received:
GET /airFlee
ESP32 Web Server: New request received:
GET /airStatus
ESP32 Web Server: New request received:
GET /CTPowerFlee
ESP32 Web Server: New request received:
GET /PowerFleePerHr
ESP32 Web Server: New request received:
GET /PowerFleePerHr
OFF
ESP32 Web Server: New request received:
GET /airStatus

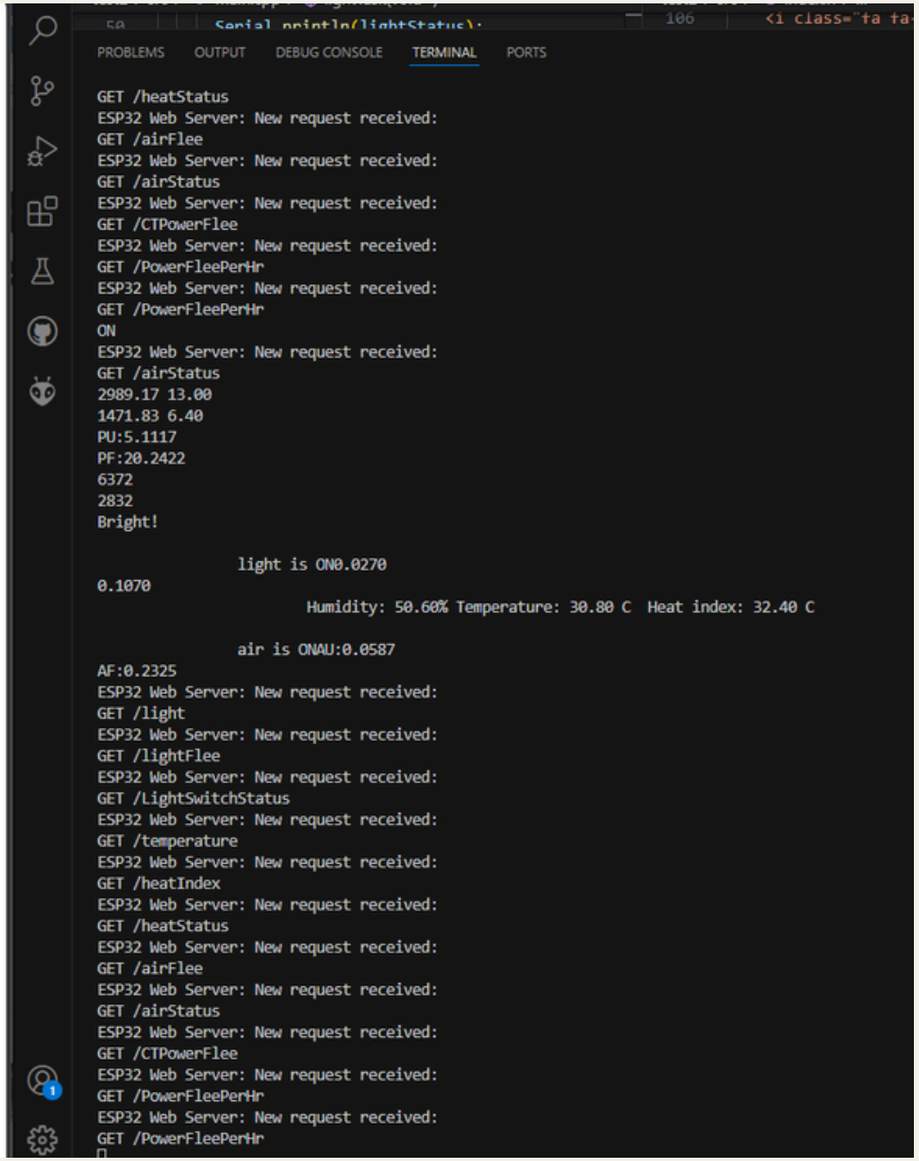
Test & Result


Home power usage monitor
Your room is now Bright!
Light switch is turn ON
Current Power Flee: 0.1047THB
Temperature: 30.70° Heat Index: 32.15°
Your room is so HOT! turn on the cooler?
Air conditioner is OFF
Air Flee: 0.2325THB
Power usage Flee: 20.0451THB
Power Flee/Hour: 19.0493THB
Power Flee/Day: 0.0000THB

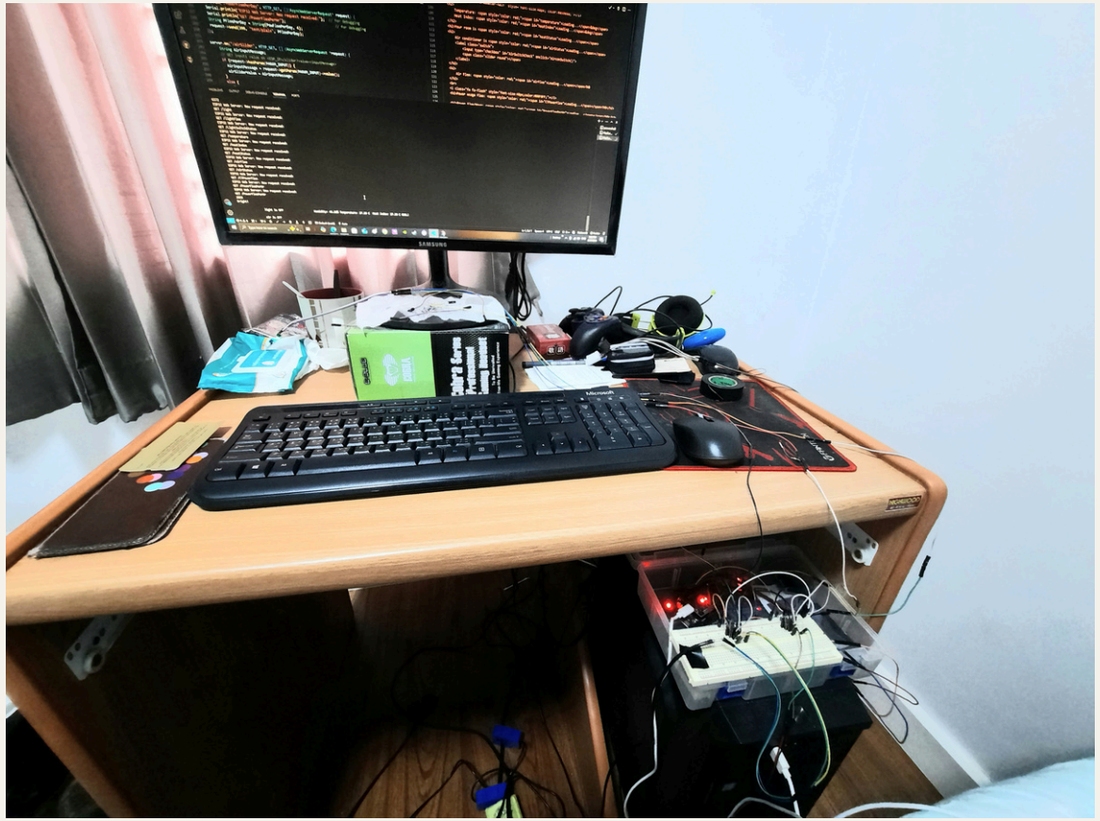

PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
0.1036
Humidity: 49.30% Temperature: 30.70 C Heat index: 32.00 C COOL!
air is OFF
1952.83 8.49
1736.56 7.55
PU:5.0619
PF:20.0451
6372
2842
Bright!
light is ON0.0264
0.1047
Humidity: 50.10% Temperature: 30.70 C Heat index: 32.15 C so HOT! turn on the co
r?
air is OFF
ESP32 Web Server: New request received:
GET /light
ESP32 Web Server: New request received:
GET /lightFlee
ESP32 Web Server: New request received:
GET /LightSwitchStatus
ESP32 Web Server: New request received:
GET /temperature
ESP32 Web Server: New request received:
GET /heatIndex
ESP32 Web Server: New request received:
GET /heatStatus
ESP32 Web Server: New request received:
GET /airFlee
ESP32 Web Server: New request received:
GET /airStatus
ESP32 Web Server: New request received:
GET /CTPowerFlee
ESP32 Web Server: New request received:
GET /PowerFleePerHr
ESP32 Web Server: New request received:
GET /PowerFleePerHr
ESP32 Web Server: New request received:
GET /light
ESP32 Web Server: New request received:
GET /lightFlee
ESP32 Web Server: New request received:
GET /LightSwitchStatus
ESP32 Web Server: New request received:
GET /temperature

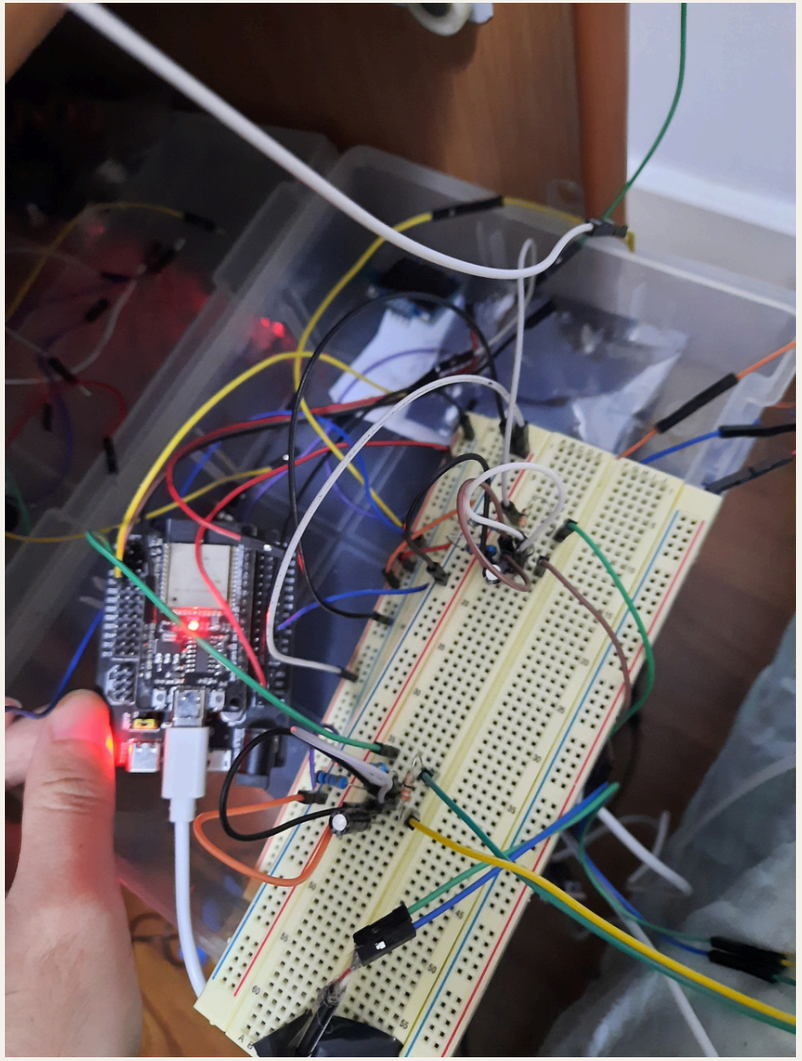
Test & Result


Home power usage monitor
Your room is now Bright!
Light switch is turn ON
Current Power Flee: 0.1070THB
Temperature: 30.80° Heat Index: 32.40°
Your room is
Air conditioner is ON
Air Flee: 0.2325THB
Power usage Flee: 20.2422THB
Power Flee/Hour: 19.0493THB
Power Flee/Day: 0.0000THB

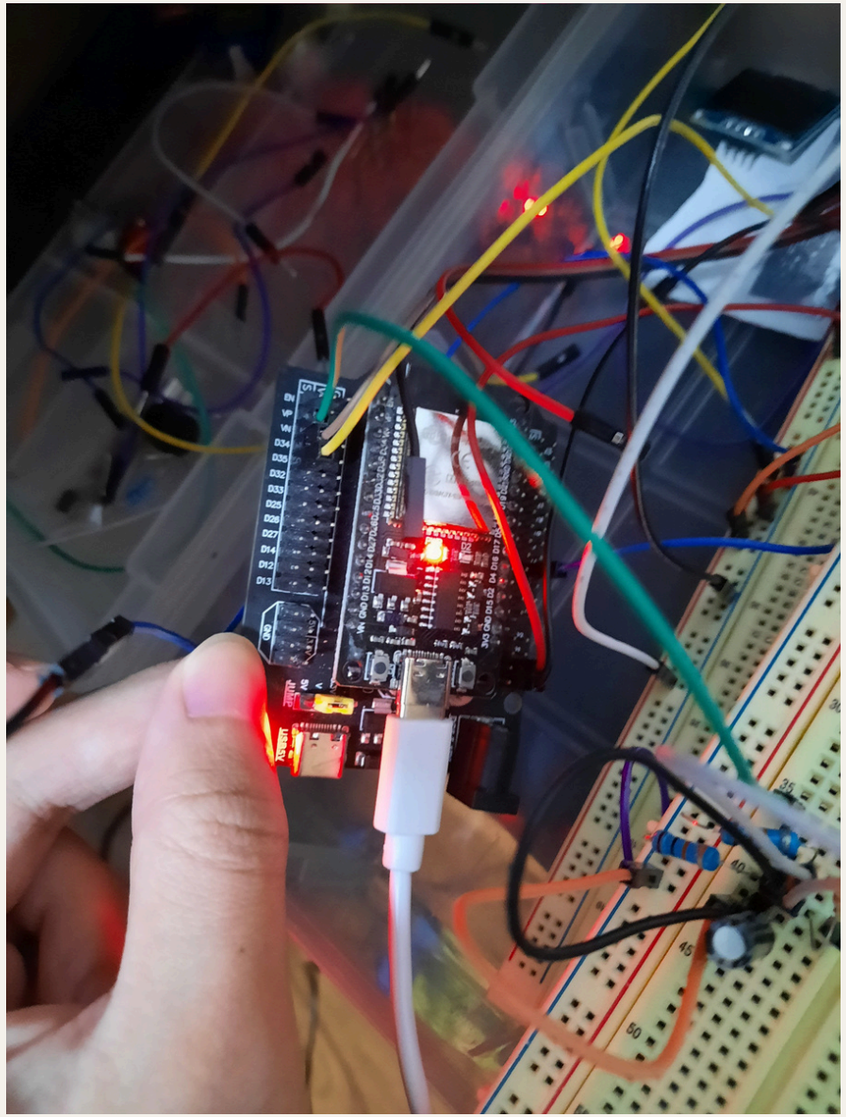

PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
GET /heatStatus
ESP32 Web Server: New request received:
GET /airFlee
ESP32 Web Server: New request received:
GET /airStatus
ESP32 Web Server: New request received:
GET /CTPowerFlee
ESP32 Web Server: New request received:
GET /PowerFleePerHr
ESP32 Web Server: New request received:
GET /PowerFleePerHr
ON
ESP32 Web Server: New request received:
GET /airStatus
2989.17 13.00
1471.83 6.40
PU:5.1117
PF:20.2422
6372
2832
Bright!
light is ON0.0270
0.1070
Humidity: 50.60% Temperature: 30.80 C Heat index: 32.40 C
air is ONAU:0.0587
AF:0.2325
ESP32 Web Server: New request received:
GET /light
ESP32 Web Server: New request received:
GET /lightFlee
ESP32 Web Server: New request received:
GET /LightSwitchStatus
ESP32 Web Server: New request received:
GET /temperature
ESP32 Web Server: New request received:
GET /heatIndex
ESP32 Web Server: New request received:
GET /heatStatus
ESP32 Web Server: New request received:
GET /airFlee
ESP32 Web Server: New request received:
GET /airStatus
ESP32 Web Server: New request received:
GET /CTPowerFlee
ESP32 Web Server: New request received:
GET /PowerFleePerHr
ESP32 Web Server: New request received:
GET /PowerFleePerHr

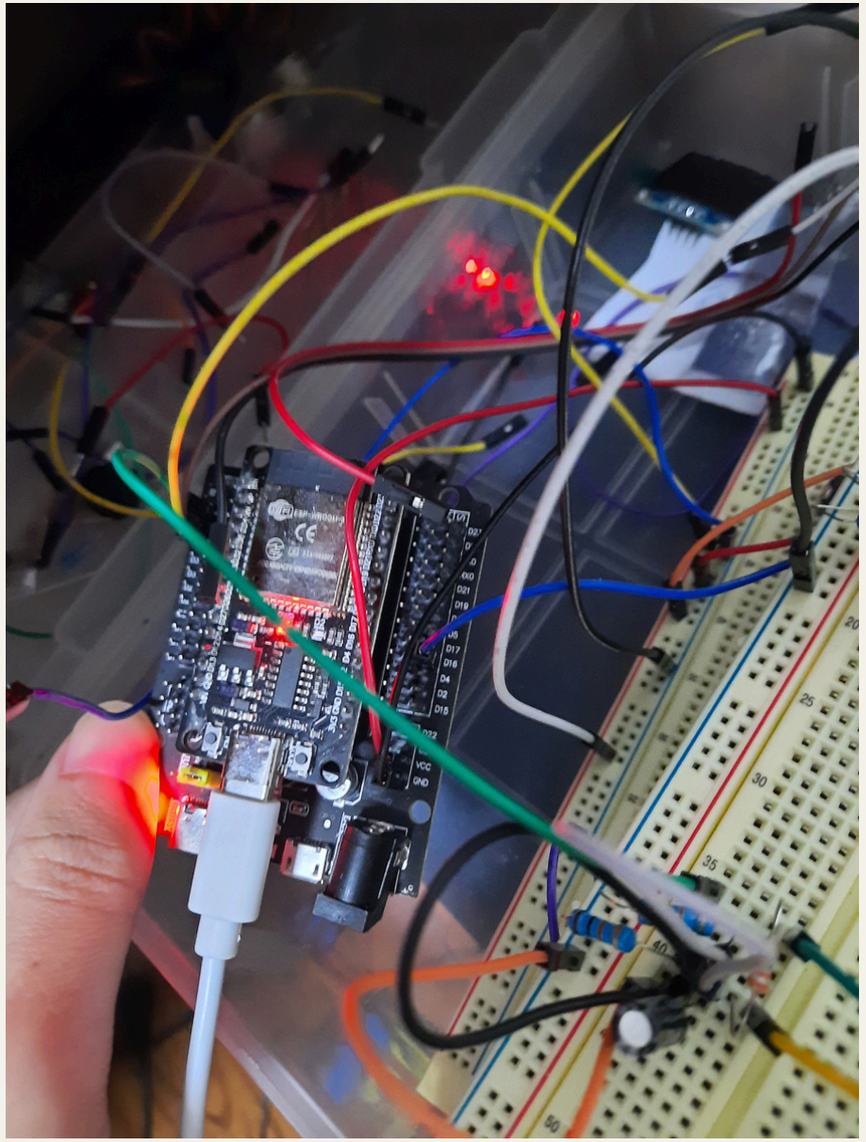
Project Picture



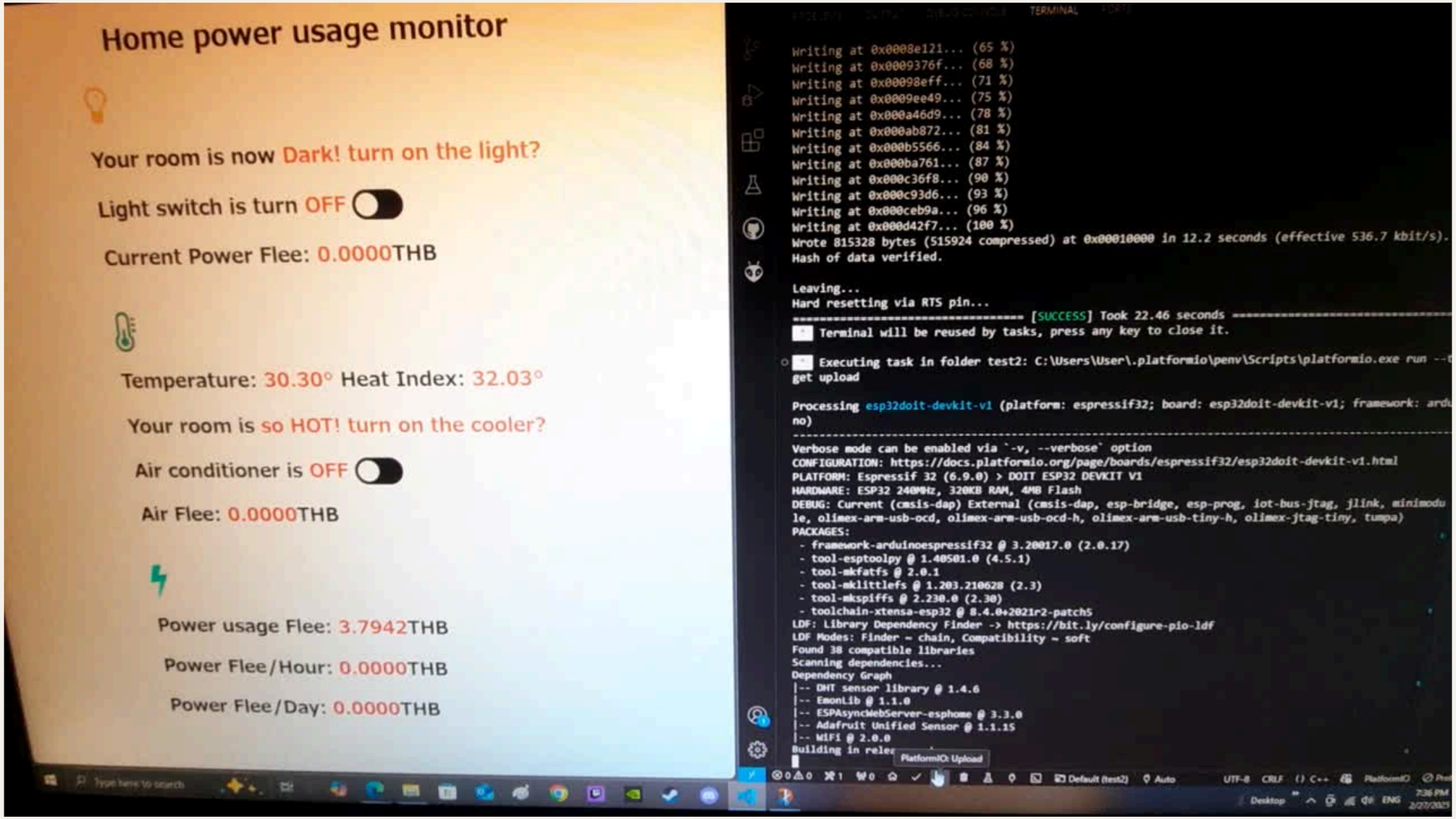


Project Picture

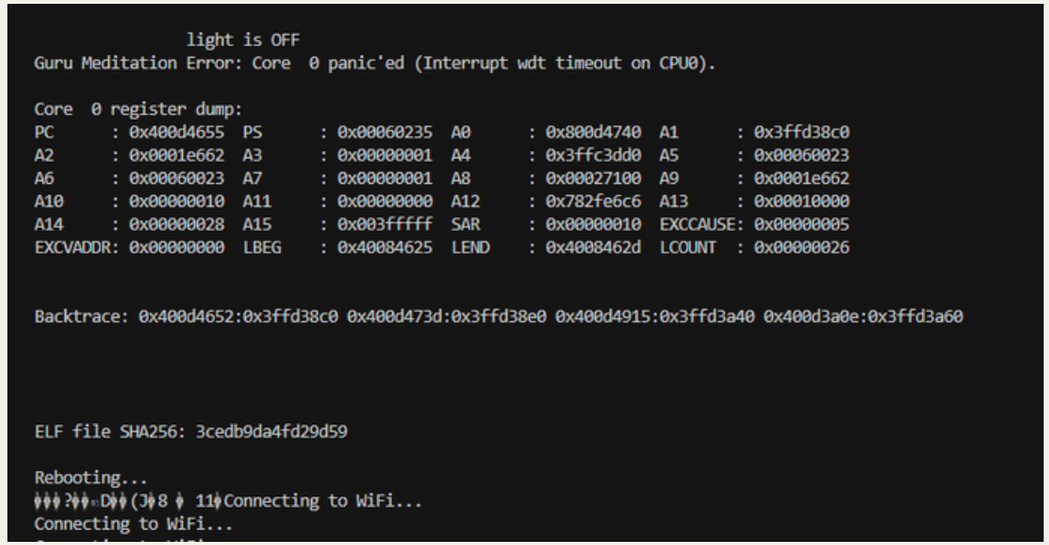




VDO Demonstration



Problem & Solution

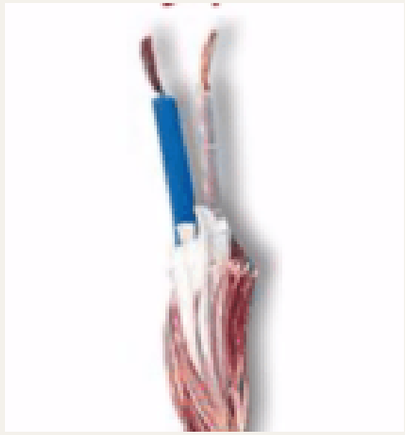


→ **use Mutex**

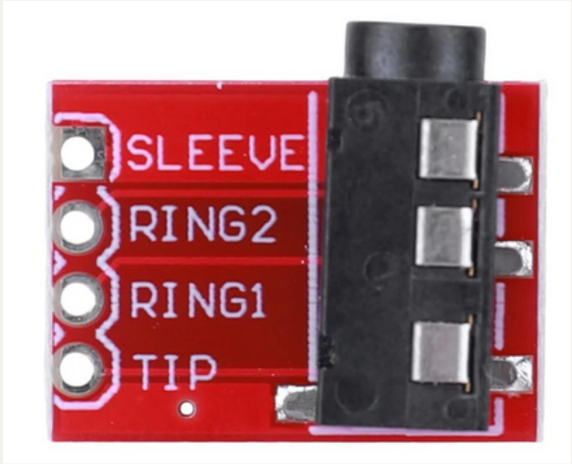
sensors can not read value when ESP32 connected to WIFI

→ **use ADC, INPUT_ONLY pin(GPIO34, 35, 36, 39)**

Problem & Solution



cut jack out



use jack breakout

Project gantt chart

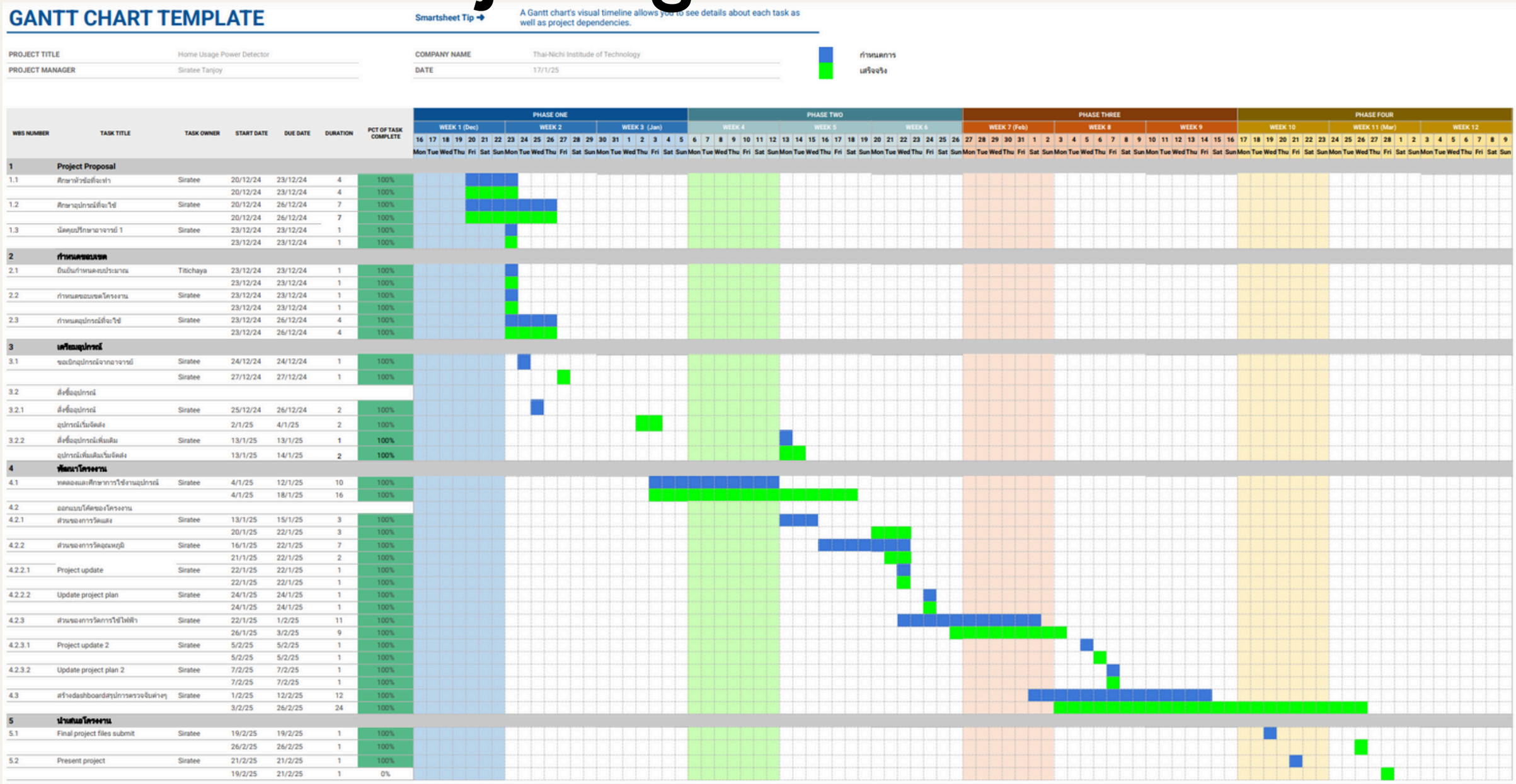
GANTT CHART TEMPLATE

Smartsheet Tip → A Gantt chart's visual timeline allows you to see details about each task as well as project dependencies.

PROJECT TITLE	Home Usage Power Detector	COMPANY NAME	Thai-Nichi Institute of Technology
PROJECT MANAGER	Siratee Tanjoy	DATE	17/1/25

กำหนดการ
เสร็จจริง

WBS NUMBER	TASK TITLE	TASK OWNER	START DATE	DUE DATE	DURATION	PCT OF TASK COMPLETE
1	**Project Proposal**					
1.1	ศึกษาหัวข้อที่จะทำ	Siratee	20/12/24	23/12/24	4	100%
			20/12/24	23/12/24	4	100%
1.2	ศึกษาอุปกรณ์ที่จะใช้	Siratee	20/12/24	26/12/24	7	100%
			20/12/24	26/12/24	7	100%
1.3	นัดคุยปรึกษาอาจารย์ 1	Siratee	23/12/24	23/12/24	1	100%
			23/12/24	23/12/24	1	100%
2	**กำหนดขอบเขต**					
2.1	ยืนยันกำหนดงบประมาณ	Titichaya	23/12/24	23/12/24	1	100%
			23/12/24	23/12/24	1	100%
2.2	กำหนดขอบเขตโครงงาน	Siratee	23/12/24	23/12/24	1	100%
			23/12/24	23/12/24	1	100%
2.3	กำหนดอุปกรณ์ที่จะใช้	Siratee	23/12/24	26/12/24	4	100%
			23/12/24	26/12/24	4	100%
3	**เตรียมอุปกรณ์**					
3.1	ขอเบิกอุปกรณ์จากอาจารย์	Siratee	24/12/24	24/12/24	1	100%
		Siratee	27/12/24	27/12/24	1	100%
3.2	สั่งซื้ออุปกรณ์					
3.2.1	สั่งซื้ออุปกรณ์	Siratee	25/12/24	26/12/24	2	100%
	อุปกรณ์เริ่มจัดส่ง		2/1/25	4/1/25	2	100%
3.2.2	สั่งซื้ออุปกรณ์เพิ่มเติม	Siratee	13/1/25	13/1/25	1	100%
	อุปกรณ์เพิ่มเติมเริ่มจัดส่ง		13/1/25	14/1/25	2	100%
4	**พัฒนาโครงงาน**					
4.1	ทดลองและศึกษาการใช้งานอุปกรณ์	Siratee	4/1/25	12/1/25	10	100%
			4/1/25	18/1/25	16	100%
4.2	ออกแบบโค้ดของโครงงาน					
4.2.1	ส่วนของการวัดแสง	Siratee	13/1/25	15/1/25	3	100%
			20/1/25	22/1/25	3	100%
4.2.2	ส่วนของการวัดอุณหภูมิ	Siratee	16/1/25	22/1/25	7	100%
			21/1/25	22/1/25	2	100%
4.2.2.1	Project update	Siratee	22/1/25	22/1/25	1	100%
			22/1/25	22/1/25	1	100%
4.2.2.2	Update project plan	Siratee	24/1/25	24/1/25	1	100%
			24/1/25	24/1/25	1	100%
4.2.3	ส่วนของการวัดการใช้ไฟฟ้า	Siratee	22/1/25	1/2/25	11	100%
			26/1/25	3/2/25	9	100%
4.2.3.1	Project update 2	Siratee	5/2/25	5/2/25	1	100%
			5/2/25	5/2/25	1	100%
4.2.3.2	Update project plan 2	Siratee	7/2/25	7/2/25	1	100%
			7/2/25	7/2/25	1	100%
4.3	สร้างdashboardสรุปการตรวจจับต่างๆ	Siratee	1/2/25	12/2/25	12	100%
			3/2/25	26/2/25	24	100%
5	**นำเสนอโครงงาน**					
5.1	Final project files submit	Siratee	19/2/25	19/2/25	1	100%
			26/2/25	26/2/25	1	100%
5.2	Present project	Siratee	21/2/25	21/2/25	1	100%
			19/2/25	21/2/25	1	0%



Special Thanks

Producer & Advisor

Titichaya Thanamitsomboon

Supplies Supporter

Jirayu Peetakul
Grisnai Chittinan
CyberTice
Arduitronics
Electronics Source
C 203

Read more...